

# Equilibrium Transition from Loss-Leader Competition: How Advertising Restrictions Facilitate Price Coordination in Chilean Pharmaceutical Retail

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## Abstract

Between December 2007 and April 2008 Chile's three retail pharmacy chains coordinated price increases on 222 medicines, weeks after advertising restrictions ended the comparative-price war that drove prices below cost. I study the transition with a demand-grounded structural model. The mechanism has two parts. *Store traffic*: comparative-price ads broadcast who is cheapest, so undercutting pays, yielding a below-cost war. *Belief*: a coordinated increase holds only if rivals expect it matched. The advertising ban moves both: by collapsing price sensitivity ( $\hat{\alpha}$  from 0.103 to 0.029) it makes undercutting unprofitable for the inelastic majority of drugs, so the coordinated price becomes a static best response, and as a public event it shifts beliefs, releasing the wave. A dynamic model estimated by simulated method of moments reproduces the path—the war, the failed attempts, and the post-ban coordination. The harm is distributional: a transfer to supra-competitive rents, with small deadweight loss because post-ban demand is inelastic.

*Keywords*: cartel formation; dynamic games; firm beliefs; loss-leader pricing; store traffic; advertising restrictions; retail pharmacy; equilibrium transition; simulated method of moments.

*JEL classification*: L41, L13, M37, L81, K21, C57.

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# 1 Introduction

Between December 2007 and April 2008 the three retail pharmacy chains that together account for 92% of pharmacy sales in Chile (Cruz Verde, Farmacias Ahumada (FASA) and Salcobrand) coordinated upward price increases on more than two hundred prescription and over-the-counter medicines. The episode was documented in detail by Chile's National Economic Prosecutor's Office (*Fiscalía Nacional Económica*, FNE) and the Competition Tribunal (*Tribunal de Defensa de la Libre Competencia*, TDLC) in two contemporaneous court records (the FNE *Requerimiento* of 9 December 2008 and the TDLC *Sentencia* N° 119/2012<sup>1</sup>) and is the setting I study here, following earlier work on collusion among retail pharmacy chains [Alé Chilet, 2016, 2018].

Through 2006 the chains discounted these medicines only occasionally; a persistent below-cost price war set in from January 2007. In August 2007 Cruz Verde launched a comparative-price advertising campaign (*Desafío*) that intensified it; in September 2007 the advertising self-regulation council (CONAR) ruled the campaign out, and on 6 November 2007 a Civil Court precautionary order upheld the restriction. The first coordinated price increase that held followed within weeks, on 3 December 2007, and the coordinated period ran to the FNE's investigation notice of 31 March 2008.

Two facts about the episode are worth separating at the outset, because they come from different sources. The first is documented: the court record shows that the chains coordinated through the drug manufacturers. A lab's *visitadores* delivered a single coordinated price proposal—a batch call—to the three chains in the same week, so the communication channel between competitors is on the record, not inferred. The second is what I add by hand-coding the coordinated price increases in the panel (§2 defines the rule: all three chains raising a given drug by at least 15% within a week, the level held): I date each attempt and separate the ones that failed from the ones that held, and among the latter a first increase that restores a 20–25% margin from a second increase to a supra-competitive level, which I call the *rent*. Three patterns follow from this coding (§3): the frequency of failed attempts is somewhat higher before and after the cartel than during it; many drugs receive two increases in sequence; and the increases arrive in batches—on a given day several products move together, and they are typically the portfolio of a single laboratory's supplier.

The transition is visible in the first batch calls (§3). Days after the 6 November court order Salcobrand raised a manufacturer's batch that its rivals did not follow, reverting within about ten days; the first batch that held came four weeks later, on 3 December, matched

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<sup>1</sup>FNE, "Requerimiento contra Farmacias Cruz Verde, Farmacias Ahumada y Salcobrand," 9 December 2008, available at [https://www.fne.gob.cl/wp-content/uploads/2011/03/requ\\_0009\\_2008.pdf](https://www.fne.gob.cl/wp-content/uploads/2011/03/requ_0009_2008.pdf); TDLC, "Sentencia N° 119/2012," 31 January 2012, available at [https://www.fne.gob.cl/wp-content/uploads/2012/01/Sentencia\\_119\\_2012.pdf](https://www.fne.gob.cl/wp-content/uploads/2012/01/Sentencia_119_2012.pdf).

by both rivals. Because the channel is explicit, that gap is the time the manufacturer-and-chains negotiation (§5) took to organize a batch all three would follow—not a learning-by-doing or tacit-coordination delay.

This article studies the transition from the loss-leader competition that preceded the coordinated period to the cartel that followed. The object is the move from one equilibrium to another, not the post-cartel price levels. How collusion is initiated, rather than how it is sustained [Green and Porter, 1984], has received less empirical attention. Two features set this episode apart from the closest work. First, the coordination is explicit: the lab-mediated channel is in the court record, so I do not infer tacit coordination from prices. Second, the build-up is fast, not gradual. Byrne and de Roos [2019], the closest empirical predecessor, documents a 2.5-year gradual rollout in retail gasoline, in which a market leader’s price experiments teach rivals their collective willingness to coordinate. This episode is a 4-month analogue. It was triggered by a single regulatory event, the 6 November 2007 Civil Court order that banned Cruz Verde’s comparative-price advertising. The onset is sharp rather than a slow build, a near-discrete flip in the static incentive rather than years of learning. The wholesale-cost record from the legal proceedings lets me identify a per-period spillover in CLP units and compute welfare across the transition.

My article contributes to the cartel-pricing literature along three related dimensions.

The first is that the transition turns on two primitives rather than one. A store-traffic value  $\mu$ , the basket profit a chain earns from the extra traffic of being cheapest, makes loss-leading rational and explains the below-cost war, since a chain prices below cost only because that traffic repays the lost margin. The prolonged war is itself evidence that this spillover exists, because sustained loss-leading would otherwise be irrational, and I estimate  $\mu$  inside the dynamic model rather than backing it out of coordinated prices. The available data cannot separate the retained wholesale margin that Cruz Verde’s integration with Socofar may have provided from pure basket spillovers, so the estimated chain-specific  $\mu_i$  pragmatically absorbs both and measures a chain’s overall resilience to the war, which is why the least-insulated chain led the escape. The second primitive is the followers’ belief that a raise will be matched: while comparative-price ads broadcast who is cheapest, undercutting pays and no chain expects a follow, so the war is a low-belief self-enforcing equilibrium, and coordination needs both the undercut incentive to vanish and the belief to shift.

The second dimension is the payoff channel through which the advertising ban, made binding by a Civil Court injunction rather than by the self-regulatory ruling that preceded it, removes the undercut incentive. By collapsing price sensitivity, the pooled  $\hat{\alpha}$  from 0.103 to 0.029, the ban shrinks the traffic a cut wins, so store traffic is no longer high enough to repay the lost margin and the one-shot undercut gain turns negative for the inelastic majority of drugs, though the most-elastic few stay positive. The coordinated price then becomes a static best response, and the wholesale-cost record from the legal proceedings

lets me measure that collapse in CLP units.

The third dimension is the belief channel, through which the ban, as a public and dated event, acts as a focal signal that shifts the expectation that a raise will now be matched, so that a regulatory event can coordinate beliefs like a commitment device rather than only changing payoffs, jolting the chains out of the low-belief war equilibrium and into the high-belief coordinated one. I embed both channels in a two-layer laboratory-to-drug dynamic game that carries the full cross-sectional heterogeneity of the coordinating drugs and is estimated by simulated method of moments, and rolled forward from the January-2007 onset it reproduces the intensifying war, the failing attempts, the post-ban wave, and the documented order in which larger laboratories coordinate first. The ban does not create the coordinated equilibrium, which is dynamically sustainable in both regimes given a high enough belief, but selects it, and the welfare cost is distributional, a transfer of about CLP 3.4bn over the cartel window with small deadweight loss because post-ban demand is inelastic.

*Related literature.* Most empirical work on collusion studies how it is sustained [Green and Porter, 1984, Harrington, 2018] or recovers collusive conduct from prices and quantities [Igami, 2017, Clark and Houde, 2014, Miller and Weinberg, 2017, Miller et al., 2021]; how a cartel begins has drawn less attention. The mechanics of reaching coordination run through communication and observable price moves: Sugar Institute communication harmonising practices [Genesove and Mullin, 2001], asymmetric gasoline adjustments acting as cartel-holding transfers [Clark and Houde, 2013], and distributor-run hub-and-spoke coordination [Chaves and Duarte, 2025]. The closest study of initiation, Byrne and de Roos [2019] (see also Byrne and De Roos, 2019), documents a two-and-a-half-year experimental rollout in retail gasoline; my episode is its short, regulatory-triggered counterpart, where one advertising restriction flips a static undercut incentive so coordination forms in four months without multi-year learning. Two studies by Alé Chilet examine this same Chilean episode—Alé Chilet [2016] on the within-cartel ordering and Alé Chilet [2018] on Salcobrand’s leadership as Bayesian signalling.

The model’s primitive is the cross-category traffic a multi-product retailer wins by being cheapest: the loss-leader logic in which below-cost “leaders” draw shoppers whose baskets repay the lost margin [Lal and Matutes, 1994, Thomassen et al., 2017, Chen and Rey, 2012, DeGraba, 2003, Florez-Acosta and Herrera-Araujo, 2020, Rao and Syam, 2001]. There store traffic sustains below-cost pricing; I run it in reverse. That restricting price advertising raises prices is an old finding, for eyeglasses [Benham, 1972] and retail pharmacy [Cady, 1976, Milyo and Waldfogel, 1999, Sinkinson and Starc, 2019], as is the broader result that an advertising ban can soften competition [Eckard, 1991]; I supply the channel: the ban leaves the store-traffic value untouched and acts through consumers’ price sensitivity, whose estimated collapse shrinks the traffic a cut wins so loss-leading stops paying.

The episode also belongs to a literature on regulation that reshapes conduct in unin-

tended ways [Carranza et al., 2015, Ryan, 2012, Cicala, 2015]. Here the consumer-protection advertising rule worked on two margins. It changed payoffs—collapsing the price sensitivity that made loss-leading pay, which made the coordinated price self-enforcing—and it shifted beliefs: a public, dated event that let the chains expect a coordinated increase to be matched. In this second role the rule acts like a focal point or commitment device, much as a price ceiling can become a focal point for tacit coordination [Knittel and Stango, 2003, Lewis, 2015]; the difference here is that the coordination is explicit and the rule released it rather than capping it. Closest is Alé-Chilet and Atal [2020]: a trade association lets many physicians coordinate the same low-to-high move; here the device is instead a public regulatory event, working through demand—a belief shift and the elasticity collapse—among only three chains.

Finally, firms often reach a new equilibrium by adaptive learning rather than jumping to it: fictitious play over three years in UK electricity [Doraszelski et al., 2018], learning-to-price after liquor privatisation [Huang et al., 2022], strategic ability in deregulation [Goldfarb and Xiao, 2011], biased beliefs in market power [Aguirregabiria and Jeon, 2020], menu-cost rigidity [Aguirregabiria, 1999, Kano, 2013]. My transition is the opposite limiting case: a dated event flips the static best response and the belief I estimate is a fast transient, so the wave is mechanical rather than learned. For the multi-product cartel structure I draw on Igami and Sugaya [2022] and for elasticity benchmarks on Grabowski and Vernon [1992]; methodologically I follow the simulated-method-of-moments tradition for dynamic games [McFadden, 1989, Pakes and Pollard, 1989, Bajari et al., 2007] and identify demand as in Berry [1994].

*Outline.* Section 2 describes the data and the regulatory record; Section 3 the descriptive patterns the model rationalises (below-cost pricing, the price-tier transition, and Salcobrand-led timing); Section 4 estimates nested-logit demand and the ban-induced collapse in price sensitivity, the pre-ban level on the competitive window and the post-ban level on the cartel-excluded weeks [Berry, 1994]; Section 5 sets out the structural model, shows it reproduces the war-to-cartel transition, and quantifies the welfare consequences; and Section 6 concludes.

## 2 Institutional Setting and Data

### 2.1 Industry and regulatory record

The Chilean retail pharmacy market in 2007 was concentrated. Cruz Verde, FASA, and Salcobrand together held approximately 92% of nationwide pharmacy sales (FNE Requerimiento 2008, chap. II)<sup>2</sup>, and were estimated to account for as much as 95% of pharmacy

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<sup>2</sup>The 92% figure is from the FNE complaint, p. 7; the 95% figure from Díaz and Galetovic [2017] is a later academic estimate covering both prescription and OTC pharmacy revenue.

revenue more broadly [Díaz and Galetovic, 2017]. Each chain set its list price for each medicine centrally at the corporate level, valid across all of that chain’s stores nationwide; the “coordination” I study in this article therefore refers to coordinated movements of the three chains’ centrally set national list prices, not to local store-level pricing.

Demand for these medicines is price-inelastic and physician-driven: the prescribing doctor, not the consumer, chooses the molecule, and a patient cannot substitute across branded products without a new prescription. The Competition Tribunal describes consumers as captive through the prescription (*cautivos a través de la receta médica*) [Tribunal de Defensa de la Libre Competencia (TDLC), 2012]. Generic competition is limited and largely outside the chains: only 22.5% of the molecules in my sample face a generic substitute, and those are sold mainly through independent pharmacies rather than the three chains, which stock predominantly branded products bought through common wholesale networks. This institutional inelasticity is the basis for the small quantity response, and hence the transfer-not-efficiency-loss welfare finding, of §5.4. This was not the first such episode: in 1993–94 the Comisión Resolutiva found the same three chains to have raised the prices of 80 ethical medicines in concert, a sanction the Corte Suprema affirmed [Comisión Resolutiva H. de Defensa de la Libre Competencia, 1995].

*Marginal-cost symmetry across the three chains.* The three chains face nearly identical wholesale-cost schedules: manufacturers publish a single suggested retail price (*precio de venta al público sugerido*, PVPS) per molecule, incorporating a 20–25% gross-margin band and sent to each chain alike, so wholesale-acquisition cost is common across chains for each drug. The one substantive asymmetry—Cruz Verde’s vertical integration with the distributor Socofar—affects margin retained at the wholesale stage, not the marginal cost of acquiring the drug. I therefore treat the wholesale cost  $c_{jt}$  as common across chains, using Salcobrand’s disclosed wholesale-cost series (§2.2) as that common  $c_{jt}$ , since the PVPS reaches the three chains alike. Chain-level *demand*, by contrast, is not forced symmetric in estimation: the demand stage (§4.4) estimates a full chain-by-drug fixed effect  $\phi_{ij}$ —a separate mean-utility intercept for each chain  $i$  on each drug  $j$ , absorbing persistent differences in how a drug sells across the three chains. The §5.1 game inherits these chain-specific intercepts  $\hat{\phi}_{ij}$ , with their overall level set so the inside-market share matches the observed  $\hat{\rho} = 0.92$  at the coordinated price, so the three chains split each drug’s nest asymmetrically as the data show. The model thus uses two firm-level asymmetries: the demand quality  $\hat{\phi}_{ij}$  just described, and a chain-specific *store-traffic* value—the basket a loss-leader draws, estimated jointly in the dynamic model and varying modestly across chains (largest for Cruz Verde, smallest for Salcobrand)—which orders the chains exactly as their below-cost exposure does and carries the Salcobrand-led escape and Cruz-Verde-led rent (§5.1).

*Origin of the price war.* The loss-leader price war that this article analyses was triggered by an advertising campaign that Cruz Verde launched in August 2007 under the slogan

“Cruz Verde Challenge: Low Prices Without Competition” (*Desafío Cruz Verde, Precios Bajos sin Competencia*). The campaign published side-by-side price comparisons for selected products against FASA, claiming systematically lower prices.<sup>3</sup> All three chains then cut prices well below cost on the products that consumers were most likely to be familiar with, to defend pharmacy traffic share. This is the loss-leader equilibrium whose collapse drives the cartel-formation episode in this article. I measure the intensity of the war by the below-cost prevalence (Fig. 1, monthly revenue shortfall from below-cost selling), which rises tenfold from June to November 2007.

*The regulatory record.* Two regulatory events bracket the period of interest. The first event is a sequence of three mutually reinforcing instruments that, between September and November 2007, terminated Cruz Verde’s comparative-advertising campaign and the loss-leader price war it had driven. On 7 **September** 2007, the board of the Advertising Self-Regulation Council (*Consejo de Autorregulación y Ética Publicitaria*, CONAR), acting on a complaint filed by FASA, issued a ruling in Case ROL 704/07 finding that Cruz Verde’s campaign infringed Articles 4, 6, 10 and 22 of the Chilean Code of Advertising Ethics (*Código Chileno de Ética Publicitaria*) and ordering Cruz Verde to cease the campaign.<sup>4</sup> CONAR, founded in 1987, is the self-regulatory organ of the Chilean National Advertisers Association (*Asociación Nacional de Avisadores*, ANDA); it adjudicates advertising-ethics complaints under that private code and its rulings are not judicially enforceable.<sup>5</sup> The board confirmed the ruling on reconsideration on 5 October 2007 and the CONAR ethics tribunal (*Tribunal de Ética*) confirmed it on appeal later in October. On 31 **October** 2007 the National Consumer Service (*Servicio Nacional del Consumidor*, SERNAC) added an administrative judicial complaint under the consumer-protection law.<sup>6</sup> The first coercive judicial instrument came on 6 **November** 2007, when the 17º Juzgado Civil de Santiago granted a *medida precautoria* requested by FASA in its parallel civil suit under Ley 20,169 (competencia desleal) and ordered Cruz Verde to discontinue the campaign, with the suit subsequently producing a definitive ruling on 2 July 2010 and a confirming sentence from

<sup>3</sup>FNE, “Requerimiento contra Farmacias Cruz Verde, Farmacias Ahumada y Salcobrand,” 9 December 2008, chap. II § 3, identifies the price war (*guerra de precios*) as launched by Cruz Verde in August 2007, with FASA and Salcobrand reacting by cutting their own prices on the same products to defend share. The press coverage in *El Mercurio*, *Economía y Negocios* (late September / early October 2007, article ID 34637) and the *Diario Financiero* of 12 September 2007 documents the campaign and FASA’s reaction.

<sup>4</sup>CONAR, Caso ROL 704/07 (“Farmacias Ahumada S.A. con Farmacias Cruz Verde S.A.”), Directorio resolution of 7 September 2007; reconsideración of 5 October 2007; and Tribunal de Ética decision later that month. The compiled jurisprudence is at [https://www.conar.cl/wp-content/uploads/2011/04/Jurisprudencia\\_Conar-2007.doc](https://www.conar.cl/wp-content/uploads/2011/04/Jurisprudencia_Conar-2007.doc).

<sup>5</sup>The Sep–Oct CONAR rulings were morally binding but not coercive, which is why the price-war prevalence kept rising through October as Cruz Verde appealed; the binding instrument was the 6 Nov Civil Court injunction documented below.

<sup>6</sup>SERNAC press communication, “Sernac denuncia a Cruz Verde por su publicidad,” 31 October 2007, available at <https://www.sernac.cl/portal/604/w3-article-918.html>.

the Court of Appeals of Santiago on 23 July 2012.<sup>7</sup> The first coordinated price increase in my hand-coded panel occurs 27 days later, on 3 **December** 2007.

The roughly thirteen-week gap from the CONAR ruling to the first coordinated increase has a natural reading: the CONAR ruling alone was not coercive; as Figure 1 (§3.1) documents, Cruz Verde’s below-cost selling intensified rather than abated in September–October 2007. The structurally consequential event was the 6 November Civil Court *medida precautoria*, which was the first instrument with the legal force to compel withdrawal of the campaign. The four-week interval between the 6 November court order and the first coordinated price increase is small relative to the typical lab-mediated coordination cycle in my empirical record (single-manufacturer batches taking multiple weeks to organise across the three chains). The structural model in §5.1 estimates the belief-shift onset  $t_0$  freely; it lands in late 2007, around the binding 6 November Civil Court injunction and the first coordinated increase on 3 December, not at the earlier non-binding CONAR ruling. Dating the public signal to the September ruling instead of the November injunction shifts the onset by eight weeks but does not alter the direction or sign of the wave.

The cartel was brought to an end by the FNE’s investigation that led to the formal complaint (*Requerimiento*) of 9 December 2008. The TDLC ultimately sanctioned all three chains, finding (i) that the chains had collusively coordinated price increases on 222 medicines, (ii) that the suggested retail margin on these medicines was 20–25%, and (iii) that the cartel’s profits represented 2.8%/4.0%/3.1% of 2007 sales for FASA, Cruz Verde and Salcobrand respectively, with the 222 coordinated medicines accounting for roughly 16% of national pharmaceutical sales (the Farma segment, 2006–2009 average).<sup>8</sup> I treat these events as exogenous in the structural model, each with its own role: the September CONAR ruling ( $t_{\text{CONAR}}$ ) is the non-binding precursor that ends the comparative campaign; the binding November ban ( $t_{\text{ban}}$ ) both collapses *demand* (the price-sensitivity drop, §4.4) and, as a public signal, shifts the common *belief* (the focal jump, §5.1); and the March FNE notice ( $t_{\text{FNE}}$ , Oficio 419) raises *enforcement* on the rent tier (the post-Oficio scrutiny penalty  $\kappa$ ), not the belief. The belief jump is dated to the binding November event, not the September ruling (§5.3).

*The batch-call coordination mechanism.* The coordination in this episode is explicit rather than tacit: pharmaceutical manufacturers in Chile distribute updated price recommendations through dedicated sales representatives (*visitadores*), who call on each chain’s central

<sup>7</sup>17º Juzgado Civil de Santiago, “Medida precautoria,” 6 November 2007 (reported by CIPER Chile, “El dossier del caso farmacias,” 9 April 2009, available at <https://www.ciperchile.cl/2009/04/09/el-dossier-del-caso-farmacias-asi-se-subieron-los-precios-segun-fasa/>); and definitive ruling of 2 July 2010, confirmed by the Court of Appeals of Santiago on 23 July 2012 (Rol 4155-2010).

<sup>8</sup>Figures from the FNE *Requerimiento* (9 December 2008) and the TDLC *Sentencia* 119/2012 (both cited above with their URLs). The profit split in (iii) is the FNE’s quantification, reproduced in the *Sentencia*; the 16% figure in (iv) is the Tribunal’s own finding (Considerando 41), measured as the coordinated medicines’ share of national pharmaceutical sales.



purchasing office on a periodic basis. Manufacturers publish a suggested retail price (*precio de venta al público sugerido*, PVPS) per drug (a list price that incorporates the standard 20–25% gross margin (§2.1)) and *visitadores* deliver revised PVPS schedules to each chain simultaneously. When a chain accepted a manufacturer’s new PVPS, it raised its prices on that manufacturer’s entire current portfolio within a single weekly window: this is a lab-batch call. The TDLC *Sentencia* 119/2012 documents that Salcobrand and rivals used the *visitadores* channel as a bilateral coordination conduit: Salcobrand would first commit to the manufacturer’s new PVPS, and the *visitador* would carry that commitment to Cruz Verde and FASA as the signal that triggered their follow-on raises.<sup>9</sup> Because the PVPS covers every presentation of a molecule from a single manufacturer, each *visitador* call naturally generates a batch across that lab’s full portfolio. Every batch call in November 2007 failed—the initiating chain raised prices and rivals declined to follow—and the first sustained three-chain coordinations emerged in the week of 3–9 December (Online Table 1 lists the principal batches).

## 2.2 Data

My primary data source is a transaction-level panel of weekly prices and quantities for 222 drugs across the three chains from 1 January 2006 through 31 December 2008. The 222 drugs are those listed in Anexo I of the FNE *Requerimiento* of 2008. The wholesale costs  $c_{jt}$  I use are those that Salcobrand disclosed to the TDLC during the legal proceedings, available for the period 1 November 2007 through 23 May 2008; I extrapolate backwards by drug-level mean for the pre-ban analysis.

*Hand-coded events.* I hand-coded the timing of every coordinated price increase from the price series, requiring that (i) all three chains raise the price of a given drug within a single 7-day window, (ii) the new level be held for at least a week, and (iii) the increase exceed 15% on the relevant chain’s list price. These criteria yield the event panel discussed in §3.3; the dynamic event ledger tracks the rent past the hand-coding’s April cutoff. Online Appendix B compares this hand-coded panel to the FNE’s official cartel indicator: the two largely agree (median timing difference –2 days; 79% of drugs within a week of each other). I use the hand-coded panel as the primary outcome throughout because it isolates the coordination event from the trailing reaction periods that contaminate the official panel.

*Enriched event panel.* I merge the 15% hand-coded ladder with a more permissive daily-panel detector ( $\geq 12\%$  spike over a prior 42-day median,  $\pm 7$ -day rival-follow window),

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<sup>9</sup>TDLC *Sentencia* 119/2012 (available at [https://www.fne.gob.cl/wp-content/uploads/2012/01/Sentencia\\_119\\_2012.pdf](https://www.fne.gob.cl/wp-content/uploads/2012/01/Sentencia_119_2012.pdf)), conduct findings; FNE *Requerimiento* 9 December 2008, chap. II, identifying the 222 coordinated medicines across 36 therapeutic categories (IMS classification). The FNE’s post-cartel market study Fiscalía Nacional Económica [2020] confirmed the *visitadores* protocol as the coordination mechanism.

classifying every detected move as a coordinated success (rivals followed within  $\pm 7$  days), a failed attempt (firm raised, rivals did not follow, firm reverted within 60 days), or a sustained-unilateral hold. This labeled panel is the source for the event counts discussed in §3. Of the 222 drugs named in the TDLC *Sentencia*, 220 (99.1%) show at least one successful three-chain coordination (Online Appendix E details the mapping and the auxiliary corporate, self-regulatory, and legal records, all publicly available with their URLs).

### 3 Descriptive Evidence

*Market structure.* Three national chains—Cruz Verde, FASA, and Salcobrand—together hold close to 90% of formal pharmacy sales and compete head-to-head for the same branded molecules, priced off a common centralized national list and bought through shared wholesale networks (§2.1). Two features of this structure shape every fact below. Demand is largely captive—the prescribing doctor picks the molecule—so the chains compete for *which* store a shopper visits, not whether she buys; this is what makes a below-cost loss leader a rational bid for store traffic. And coordination, when it comes, runs through the manufacturers’ batch calls rather than chain-to-chain, so the unit that moves is the laboratory’s portfolio, not the single drug.

This section documents, with no recourse to any estimated parameter, the facts that are the descriptive counterpart of the structural argument the rest of the article makes. Against the regulatory timeline set out in §2.1 (the war that Cruz Verde’s August-2007 campaign drove, the CONAR ruling and 6 November Civil Court order that ended it, and the FNE investigation that closed the cartel), the facts read together describe a loss-leader equilibrium whose intensity was accelerating through 2007 toward a level that could not be sustained, and a coordinated wave of price increases that restored the chains’ weighted pharmacy margin to roughly the historical 20–25% band. I establish three facts, each with no recourse to an estimated parameter, and organise the section around them: (i) pre-ban below-cost pricing whose intensity was accelerating, not stationary (§3.1); (ii) a sharp price-tier transition that restored the historical margin and was not a cost shock (§3.2); and (iii) Salcobrand-dominant leadership and a four-month wave organised laboratory-by-laboratory, the largest laboratories first (§3.3). Sections 4–5.4 build the structural model that rationalises the magnitudes; this section establishes that they are there to be explained.

#### 3.1 Fact 1: Below-cost pricing accelerated before the ban

The standard formal argument for a loss-leader equilibrium is that firms expect the per-period loss on the loss-leader to be recovered from contemporaneous spillovers on other categories (§5.1). A natural diagnostic of whether such an equilibrium is sustainable is

whether the loss is approximately stationary or whether it grows over time. The below-cost gap is visible directly in Figure 1: across 2007 each chain’s quantity-weighted price (Panel B) runs below the wholesale cost (Panel A), and the gap widens through the year. Panel C aggregates, for each chain, the monthly revenue shortfall from below-cost sales: that is,  $\sum_{j,t} \max(c_j - p_{ijt}, 0) \cdot q_{ijt}$  summed across drugs and days within each month.<sup>10</sup>

Through 2006 and the first half of 2007, the monthly shortfall is roughly steady at ~50–100 million CLP per chain. From mid-2007 onwards, every chain’s shortfall begins to grow, and the growth accelerates into November–December 2007, with Cruz Verde reaching a peak of roughly ~900 million CLP per month. The chains were bleeding roughly an order of magnitude more in late 2007 than they were a year earlier. Within weeks of the first coordinated increase, the shortfall is back to baseline.

This pattern is hard to reconcile with the loss-leader equilibrium having been “stable” through 2007. It is strong descriptive evidence for the thesis the structural model formalises and motivates the structural measurement: the firms moved as the net payoff to staying put in the war deteriorated.

### 3.2 Fact 2: The price-tier transition

Figure 1 (Panel B) shows the weekly quantity-weighted average price for each chain over the 2006–2008 window. Three features stand out. First, prices co-move throughout: across the 156 weeks of data, the three chains are within roughly  $\pm 500$  CLP of one another at the weekly frequency, consistent with the centralised national list-pricing described in §2.1. Second, prices drift downward from mid-2006 through late 2007 (from ~7,000 to ~5,000 CLP), with the dip steepening between the CONAR ruling and the first coordinated increase. Third, in a single sharp step in December 2007–February 2008, the chain-level price returns to and exceeds the 2006 level. This step-up was not a cost shock: over the same coordination window the chains’ wholesale costs rose only about 3.3% (Salcobrand’s WAC series disclosed to the Competition Tribunal, December 2007–May 2008; +2.2% over the longer November–May window of Figure 1), against retail increases of 28–60% across therapeutic categories [Tribunal de Defensa de la Libre Competencia (TDLC), 2012]. The reversal reflects coordinated conduct, not input costs.

The coordinated step restores a sustainable, nonnegative margin: quantity-weighted markups move from substantially negative (–12 to –16%) at the war floor to small positives at the first ( $\ell = 0 \rightarrow 1$ ) step—still below the 20–25% suggested-retail band the TDLC documents (*Sentencia* 119/2012, Cap. XI)—while a second push ( $\ell = 1 \rightarrow 2$ ) overshoots to

<sup>10</sup>The cost benchmark  $c_j$  is the drug’s wholesale cost from the Salcobrand cost series, which the data record from November 2007; for earlier months I hold  $c_j$  at its first observed level, since the wholesale price of these off-patent molecules is stable over the window. Because the pre-November benchmark is backward-extrapolated, the *level* of the early shortfall is measured with more error than the post-November level; the acceleration into late 2007 is robust to this, but its exact onset month should be read as approximate.

+34 to +41%, into the region the Tribunal flags as supra-competitive. This asymmetry maps onto the structural model directly: the first step is the regime flip (the ban turns the one-shot undercut gain  $G_j$  negative, restoring a sustainable margin), the second a selective rent-extraction step taken by only the most inelastic drugs (§5). Online Appendix Table 4 decomposes the markup by chain and tier.

The movements are driven by prices, not costs. Over the war-trough-to-cartel window (November 2007–May 2008) Salcobrand’s wholesale-cost index—the WAC series submitted as expert evidence—rises just +2.2% (per-drug coefficient of variation  $\approx 1\%$ ), while the quantity-weighted retail price rises about 43% (Figure 1, Panel A). A cost shock cannot account for a price path that fell during the war and then rose far more than costs during the recovery; the driver is the demand-side regime change at the advertising ban.

### 3.3 Fact 3: Leadership and timing

Salcobrand led 247 of the 363 coordination events (68%) and dominated the cartel wave (211 of 284 coordinations), even though Cruz Verde bore the heaviest pre-ban below-cost exposure (31.3% of drug-weeks below cost against Salcobrand’s 25.6%, FASA between). This is the signature of the store-traffic mechanism in §5: a chain goes deeper below cost the more basket profit its loss-leader draws, so Cruz Verde’s heaviest exposure marks it the highest-store-traffic chain and Salcobrand’s lightest marks it the lowest—and the lowest-traffic chain, hurt most by the war, has the strongest incentive to escape it and leads. The case record confirms the order—the FASA minute notes that Salcobrand “very habitually” raised first (*muy habitualmente Salcobrand*; Tribunal de Defensa de la Libre Competencia (TDLC), 2012)—and in the post-investigation rent Cruz Verde, the comfortable high-traffic aggressor, overtakes it (36 initiations to Salcobrand’s 34), the SB  $\rightarrow$  CV handoff the model reproduces. The estimated store-traffic ordering matches the below-cost exposure exactly, an out-of-sample check the leadership shares do not pin. Online Appendix Table 5 reports the full leadership split.

The wave ran 3 December 2007 to 31 March 2008: 216 of the 220 first coordinations (Tier 0  $\rightarrow$  1) fall within these 19 weeks, while the rent steps (Tier 1  $\rightarrow$  2) ratchet on through year-end. Over the full episode the chains made 220 first coordinations and 143 rent steps—the 363 events the structural model targets, 284 of them in the cartel-window success moment—against only 24 salient failed or unilateral attempts (a 7.6% failure rate; the model’s finer count of every reverted attempt is 194); 228 of the 363 coordination steps exceed 25%. The ordering was organized at the laboratory level: each manufacturer’s visitador delivered a batch across its portfolio, so timing is set by which laboratory engaged first. Larger laboratories coordinated first—a one-SD increase in log pre-ban revenue raises the coordination hazard by 25% (Cox HR = 1.25,  $z = 3.3$ )—and lab membership alone explains 57% of the cross-drug timing variance, against 3% for a drug’s

own size, elasticity, and below-cost depth (Online Appendix Table 6). Before the cartel held, the chains were already testing: 35 attempts from 6 November to 2 December failed (Figure 4), the largest a Salcobrand-led 17-drug Laboratorio Chile batch on 9–13 November that neither rival followed.

Figure 2 shows the size gradient directly: a laboratory’s median first-coordination week falls with its size—the log of its total pre-ban revenue, the same stake  $W_L$  the structural model assigns it—at a correlation of  $-0.29$ . Table 1 records the principal batch calls: every batch call in October and November failed, and the first sustained three-chain coordinations emerged in the week of 3–9 December.

Figure 3 illustrates one episode. The weekly price for Sildenafil Citrate (Drug #80) drifts down through mid-2007, then steps up within a single week in early December 2007 (tier  $0 \rightarrow 1$ , led by Salcobrand) and again in early April 2008 (tier  $1 \rightarrow 2$ ); the three chains co-move tightly across both transitions. The official binary indicator  $col_{ijt}$  records only the first step (it is monotone per drug), so it understates the cartel’s reach by about one-third—records fewer steps than my coding in 127 of the 137 multi-round drugs (Online Appendix B)—which is why I use the hand-coded tier ladder throughout.

The failure rate (Figure 4) traces the regime:  $\approx 97\%$  through the first four weeks of November, below 10% by mid-December, near zero through March, then a snap back after the FNE issues Oficio N° 419 on 31 March 2008, when new coordination stops while the cartel holds and the rent deepens. The reverted lifts in this last window are failed new attempts on not-yet-coordinated drugs, not defections: the 220 members hold their coordinated price throughout (Online Appendix Table 7 gives the full ledger by regime window and magnitude). Salcobrand leads the failed re-coordination push (82 of 127 single-chain lifts), the same war-exposed chain (the lowest- $\mu$  Salcobrand) that led the escape from it.

The dynamic structural model of §5 takes this loss-leader regime as the period-0 state and asks who exits it and when; the next three sections build that model.

## 4 Demand

### 4.1 Model

A consumer in week  $t$  decides whether to fill a prescription for drug  $j \in \{1, \dots, J\}$  at one of the three national chains  $i \in \mathcal{I} = \{\text{CV}, \text{FA}, \text{SB}\}$ , or to use the outside option (an independent pharmacy or no purchase). I use a two-level nested logit in which the three chains form a single inside nest for each drug and the outside option is the only competing

nest.<sup>11</sup> Consumer  $h$ 's indirect utility from buying drug  $j$  at chain  $i$  is

$$U_{hijt} = u_{ijt} + v_{hjt} + (1 - \sigma) \varepsilon_{hijt}, \quad \sigma \in [0, 1), \quad (1)$$

where  $v_{hjt}$  is the nest-level taste shock common to the three chains,  $\varepsilon_{hijt}$  is an i.i.d. Type I extreme-value shock, and the nesting parameter  $\sigma$  measures within-nest correlation across the three chain alternatives. Let  $D_t \equiv \mathbf{1}\{t \geq t_{\text{CONAR}}\}$  be the demand-regime indicator, switched at the September CONAR ruling that opens the restricted-advertising regime; the structural model dates the binding collapse in price sensitivity, together with the belief shift it brings, to the November court ban that made the restriction enforceable (§2.1). Mean utility is

$$u_{ijt} = X_{jt}\beta - (\alpha_0 + \alpha_1 D_t) p_{ijt} + \phi_{ij} + \xi_{ijt}, \quad \alpha_0 > 0, \quad \alpha_0 + \alpha_1 > 0, \quad (2)$$

where  $p_{ijt}$  is the transaction price (CLP 000s),  $\phi_{ij}$  is a chain-by-drug fixed effect,  $X_{jt}\beta$  collects time-varying drug characteristics, and  $\xi_{ijt}$  is an unobserved demand shock. The price coefficient is decomposed into a pre-ban level  $\alpha_0$  and a post-ban shift  $\alpha_1$ ; the null of no ban effect on price sensitivity is  $H_0: \alpha_1 = 0$ . The sign convention  $\alpha_0, \alpha_0 + \alpha_1 > 0$  makes (2) consistent with downward-sloping demand on both sides of the ban.<sup>12</sup>

Aggregating over consumers yields the within-nest share  $s_{i|jt}$ , the nest share  $s_{jt} = \sum_{i \in \mathcal{I}} s_{i|jt}$ , and the outside share  $s_{0jt} = 1 - s_{jt}$ . Taking logs and differencing relative to the outside good [Berry, 1994] gives the single pooled estimating equation

$$\underbrace{\ln s_{ijt} - \ln s_{0jt}}_{Y_{ijt}} = X_{jt}\beta - \alpha_0 p_{ijt} - \alpha_1 (D_t \cdot p_{ijt}) + \sigma \ln s_{i|jt} + \phi_{ij} + \xi_{ijt}. \quad (3)$$

Equation (3) is estimated in two passes. The pre-ban pass is a single GMM system on the pooled *weekly* panel—weekly is the natural prescription-refill unit and the standard frequency in pharmaceutical demand studies [Ellison et al., 1997, Björnerstedt and Verboven, 2016]—identifying the market-level objects, the nesting  $\sigma$  and the pooled sensitivity  $\alpha_0$ , from the cross-section with daily noise smoothed out; the interaction  $D_t \cdot p_{ijt}$  returns the ban shift  $\hat{\alpha}_1 = -0.074$ . The post-ban sensitivity  $\hat{\alpha}^{\text{post}} = 0.029$  is re-estimated separately on the cartel-excluded competitive weeks—the rival-price instruments fail inside the cartel window, where coordinated prices co-move with the demand shock—holding  $\sigma$  at its pre-ban value so only the price coefficient moves, which isolates the ban-induced collapse. The per-drug coefficients  $\hat{\alpha}_j$  instead use the daily panel, whose far larger count of within-drug,

<sup>11</sup>Drug  $j$  is defined at the molecule-dosage level; the three chains carry the same brand-package portfolio for each  $j$  (Section 2). Prescription pre-selects  $j$ , so substitution across drugs is absorbed in the outside good.

<sup>12</sup>Throughout, I report estimates of  $\alpha_0$  and  $\alpha_0 + \alpha_1$  as the relevant price sensitivities;  $\alpha_1$  is the ban-induced shift.

cross-chain price observations gives the stronger first stage a drug-specific slope needs.

The market size for drug  $j$  is

$$N_j = \frac{\max_t Q_{jt}^{\text{wk}}}{\hat{\rho}} + 1, \quad (4)$$

where  $Q_{jt}^{\text{wk}} = q_{\text{CV},jt} + q_{\text{FA},jt} + q_{\text{SB},jt}$  is the total weekly three-chain quantity and  $\hat{\rho} = 0.92$  is the three-chain share of Chilean pharmacy sales (FNE Requerimiento 2008, p. 7; Díaz and Galetovic, 2017 estimate  $\hat{\rho} = 0.95$ ). Dividing by  $\hat{\rho}$  calibrates the outside good to represent consumers at independent pharmacies ( $\approx 8\%$  of national sales) plus eligible non-buyers in a given week. At the peak week,  $s_{0j} \approx 8\%$ ; at an average week,  $s_{0j} \approx 37\%$ . The constant  $+1$  prevents a degenerate  $s_{0jt}$  in the peak observation; the chain-by-drug fixed effect  $\phi_{ij}$  absorbs any remaining level mis-scaling.<sup>13</sup>

The baseline specification imposes a single  $\sigma$  and a single pre-ban price coefficient  $\alpha_0$  rather than allowing each across products or treatment categories: with three chains per drug-week, in-sample variation does not separately identify drug-specific price coefficients from  $\phi_{ij}$ . The ban-induced shift  $\alpha_1$  is identified off the price–quantity comovement in the post-ban window. The cross-product dispersion in implied elasticities reported in Section 4.5 therefore reflects variation in prices and shares, not preference heterogeneity. The homogeneity restriction is itself an empirical question; I relax it in Section 4.4 below and, as a per-molecule short-run Salop robustness, in a companion technical memo.

## 4.2 Identification and Instruments

Two regressors in (3) are endogenous: price  $p_{ijt}$  (correlated with  $\xi_{ijt}$  through unobserved demand shocks) and the within-nest share  $\ln s_{iljt}$  (correlated with  $\xi_{ijt}$  mechanically). The instrument set is four variables:

1. **Wholesale cost (drug average)**,  $\bar{c}_j$ : the weekly average of Salcobrand’s recorded wholesale cost for drug  $j$ , averaged across weeks. Drug-level cost variation pins down the cross-section of  $\alpha$ .
2. **Wholesale cost (weekly)**,  $c_{jt}$ : Salcobrand’s wholesale cost in week  $t$ . Within-drug, week-to-week cost variation pins down the time-series identification of  $\alpha$ .
3. **Rival price 1**,  $p_{-1,jt}$ , the contemporaneous price of one rival chain.
4. **Rival price 2**,  $p_{-2,jt}$ , the price of the other rival chain.

<sup>13</sup>I have verified robustness to two alternative calibrations: (i)  $\hat{\rho} = 0.95$  (Díaz-Galetovic) and (ii) a constant national population  $N^*/J$  (see Online Appendix D);  $(\hat{\alpha}, \hat{\sigma})$  shift by less than  $\pm 0.01$  in either case.

The two cost instruments shift the marginal cost faced by all three chains symmetrically (the wholesale market is national), so they enter  $p_{ijt}$  but, conditional on  $\phi_{ij}$  and the time controls in  $X_{jt}$ , are excluded from  $\xi_{ijt}$ . The two rival prices follow the Hausman [1996] logic: they shift the within-nest share  $\ln s_{iljt}$  through demand substitution but are otherwise excluded. I do not use lagged own prices or lagged shares.<sup>14</sup>

*Instrument validity across the ban.* The two cost instruments are excluded in every regime. The two rival-price (Hausman) instruments are valid only when the chains compete: once they coordinate, rival prices co-move with the demand shock  $\xi_{ijt}$  and are no longer excluded. The ad ban does not break the instruments—the cartel does. Pooling all ads-banned weeks the over-identifying  $J$  rejects ( $J = 14.5$ ,  $p = .001$ ), but the rejection is confined to the collusive cartel window (3 Dec 2007–11 Apr 2008): dropping those weeks and re-estimating on the competitive ads-banned weeks, the same four instruments pass the over-identifying test ( $J = 4.4$ ,  $p = .11$ ; Table 2, note ‡). I therefore identify  $\alpha_0$  on the pre-ban weeks and the post-ban price sensitivity on the cartel-excluded competitive weeks, with  $\sigma$  pooled at its pre-ban value 0.393 so that price is the only endogenous regressor. This yields  $\hat{\alpha}^{\text{post}} = 0.029$ ; the cost-only IV (0.033) and the post-cartel-only window (0.028) corroborate it without using the rival prices at all.

### 4.3 Estimation and Test Statistics

I estimate (3) by two-step IV-GMM [Berry, 1994, Conlon and Gortmaker, 2020] on the pooled chain-drug-week panel, absorbing the chain-by-drug effect  $\phi_{ij}$  with a within-group transform (instruments demeaned likewise). The moment condition is  $E[Z_{ijt} \xi_{ijt}(\theta)] = 0$  with  $\theta = (\alpha_0, \alpha_1, \sigma, \beta)$ ; each instrument enters at both its direct and ban-interacted level  $D_t \cdot Z_{ijt}$ , which separates  $\alpha_0$  from  $\alpha_1$  (§4.2). Standard errors treat  $\xi_{ijt}$  as heteroskedastic across  $(i, j, t)$ ; drug-level clustering is a robustness check.

On each window I report the first-stage  $F$ -statistics for  $p_{ijt}$  and  $\ln s_{iljt}$  [Stock and Yogo, 2005], the Hansen [1982] overidentification  $J$  (degrees of freedom  $q - k = 1$  on the pre-ban window, where  $\sigma$  is jointly estimated, and  $q - k = 2$  on the post-ban windows, where  $\sigma$  is held at its pre-ban value so price is the only endogenous regressor), and the fitted–observed correlation in  $Y_{ijt}$  (the nested-logit analogue of  $R^2$ ). I impose no bounds on  $\alpha$  or  $\sigma$ .

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<sup>14</sup>The previous draft used 118 post-LASSO lagged-price/share instruments. Lagged shares are mechanically correlated with  $\xi_{ijt}$  under serially dependent demand shocks, and lagged-price IVs collapse to OLS in fixed-effect specifications because prices are sticky. The current set discards both.



#### 4.4 Estimates

Table 2 reports the demand estimates. The headline row is the pooled Model B result ( $\hat{\alpha}_0 = 0.103$ ,  $\hat{\sigma} = 0.393$ ; cartel-excluded post-ban  $\hat{\alpha}^{\text{post}} = 0.029$ ) that the structural model consumes; the lower block reports the per-drug coefficients  $\hat{\alpha}_j$  that give the model its drug-level heterogeneity. The headline pooled row (Panel A) uses the weekly panel and the weekly market-size calibration (4); the per-drug coefficients (Panel B) use the daily panel ( $T = 1,096$  days, 222 drugs) with the daily analogue  $N_j = 7 \cdot \max_t Q_j^{\text{daily}} / \hat{\rho} + 1$  (the factor of seven converts peak-day quantity to a weekly-equivalent market).

**Per-drug coefficients.** Holding the nesting parameter at the pooled  $\hat{\sigma} = 0.393$ , I estimate the per-drug  $\hat{\alpha}_j$  for each of the 222 drugs by two-stage least squares on the competitive (cartel-excluded) daily panel ( $\approx 3,288$  obs per drug across three chains), instrumenting each drug’s price (and its post-cartel interaction) with the two rival-chain prices.<sup>15</sup> The within-transform removes chain (Firm) effects only: a two-way Firm $\times$ Day demeaning would force the three chains’ prices to sum to zero within each day, making own price a mechanical function of the rivals and the rival instrument degenerate. The within-drug estimating equation is

$$\tilde{Y}_{ijt} = -\alpha_j P_{ijt} - \gamma_j (P_{ijt} \times \text{PostCartel}_t) + \phi_i + \varepsilon_{ijt}, \quad (5)$$

with  $\tilde{Y}_{ijt} = Y_{ijt} - \hat{\sigma} \ln s_{ijt}$ ,  $\text{PostCartel}_t = 1$  for days 829–1,096,  $\phi_i$  a chain fixed effect, and  $P_{ijt}$  (and its interaction) instrumented by the rival prices. Identification of  $\alpha_j$  relies on within-drug, cross-chain price dispersion in the competitive windows. The ban interaction  $\gamma_j$  enters only as a control, so  $\hat{\alpha}_j$  is the clean pre-ban within-window price slope; the post-ban  $\hat{\alpha}_j$  reported below is re-estimated with the Salcobrand-cost instrument, since post-ban rival prices are coordination-contaminated (note ‡), and the structural model scales each  $\hat{\alpha}_j$  to its post-ban value by the common pooled ratio  $\alpha^{\text{post}}/\alpha_0 = 0.29$  (§5). The pooled analog of the ban shift is  $\hat{\alpha}_1 = -0.074$ .

<sup>15</sup>The nesting parameter  $\sigma$  governs the single chain nest—the three chains compete for the same drug—so it is a market-level, not a drug-level, object. I therefore identify it once on the pooled pre-ban data (Panel A) and condition the per-drug slopes on it; a separate per-drug  $\sigma_j$  is not identified, as it would need within-drug substitution variation that three chains over a short window cannot supply. Estimating a common  $\sigma$  with heterogeneous price coefficients  $\alpha_j$  is the standard sequential nested-logit approach [Berry, 1994]. Online Appendix Table 12 (Panel B) re-solves the structural model at  $\sigma \in \{0.30, 0.50\}$ : the  $\mu$  band and the war-to-coordination flip are unchanged, so the fixed- $\sigma$  choice does not drive the results.

Table 2: Heterogeneous price coefficient by drug class

**Panel A. Pooled headline** (homogeneous  $\alpha$ ,  $\sigma$  pooled at the pre-ban value)

|                                          | $\hat{\alpha}$          | $\sigma$         | $N$    | $J(p)$      |                                              |
|------------------------------------------|-------------------------|------------------|--------|-------------|----------------------------------------------|
| Pre-ban (cost + rival IV)<br>SE          | 0.103<br>(0.005)        | 0.393<br>(0.107) | 62,985 | 0.60 (.44)  | identifies $\alpha_0, \sigma$                |
| Post-ban, all weeks (incl. cartel)<br>SE | 0.022<br>(0.001)        | [0.393]          | 41,769 | 14.5 (.001) | not used <sup>‡</sup>                        |
| Post-ban, cartel-excluded<br>SE          | <b>0.029</b><br>(0.001) | [0.393]          | 29,172 | 4.4 (.11)   | used; $\alpha^{\text{Post}}/\alpha_0 = 0.29$ |

**Panel B. Per-drug  $\hat{\alpha}_j$  by drug type:** 2SLS, Firm FE, cartel weeks dropped, pooled  $\hat{\sigma} = 0.393$ ; pre-ban rival-price IV, post-ban Salcobrand cost *and* rival IVs (daily panel)

| Drug type            | $n_{\text{drugs}}$ | Pre-ban $\hat{\alpha}_j$ | Post-ban $\hat{\alpha}_j$ | post/pre |
|----------------------|--------------------|--------------------------|---------------------------|----------|
| Chronic $\times$ Rx  | 130                | 0.123                    | 0.023                     | 0.19     |
| Chronic $\times$ OTC | 66                 | 0.127                    | 0.061                     | 0.48     |
| Acute $\times$ OTC   | 11                 | 0.229                    | 0.026                     | 0.12     |
| Acute $\times$ Rx    | 15                 | 0.214                    | n.i.                      | —        |
| All 222 drugs        | 222                | 0.130                    | 0.029                     | 0.22     |

<sup>‡</sup> On all ads-banned weeks the over-identifying  $J$  rejects ( $J = 14.5$ ,  $p = .001$ ): the cartel weeks (3 Dec 2007–11 Apr 2008) are collusive, so coordinated rival prices co-move with the demand shock and the Hausman instruments fail exclusion. Dropping them restores validity ( $J = 4.4$ ,  $p = .11$ ) and identifies  $\hat{\alpha} = 0.029$  ( $\sigma$  held at the pre-ban 0.393), the post-ban demand the structural model (§5) consumes ( $\alpha^{\text{Post}}/\alpha_0 = 0.29$ ). A cost-only IV (0.033) and the post-cartel window (0.028) corroborate it; the flip and the  $\mu^*$  band are robust across [0.028, 0.033].

*Per-drug notes.* Pre-ban  $\hat{\alpha}_j$  is the per-type median of each drug's competitive slope from (5): per-drug 2SLS with price instrumented by the two rival-chain prices within Firm fixed effects, valid under competition (median first-stage  $F = 1,324$ ; 97% with  $|t| > 1.96$ ; the 5 drugs with  $\hat{\alpha}_j \leq 0$  take the type mean;  $\alpha$ 's are absolute values). Post-ban  $\hat{\alpha}_j$  adds the Salcobrand wholesale-cost instrument to the rival prices—the full valid IV set on the cartel-excluded sample, since the rival prices alone are coordination-contaminated in-cartel (note ‡);  $\hat{\sigma}$  pooled at 0.393, HC0 errors. The cost series begins Nov 2007, so the post-ban coefficient is identified on the competitive Nov-2007 and post-cartel weeks: three of the four classes collapse cleanly to 0.023–0.061 and the all-drug estimate is 0.029, matching the headline; only the 15-drug Acute $\times$ Rx class is too thin to identify separately (n.i.). The structural model scales each pre-ban  $\hat{\alpha}_j$  by the pooled ratio 0.29.

The per-drug coefficients line up with prior knowledge of pharmaceutical demand. Acute-condition drugs are more price-sensitive than chronic-condition drugs (median  $\hat{\alpha}_j$  of 0.23 for Acute $\times$ OTC vs. 0.12 for Chronic $\times$ Rx), matching the intuition that refill prescriptions create switching costs that dampen price response. Splitting each type's coefficient into its pre- and post-ban value (Panel B) shows the ad-ban collapse at the

drug level: with the cost and rival instruments, three of the four classes fall to 0.023–0.061 and the all-drug estimate to 0.029, matching the pooled headline, while only the 15-drug Acute×Rx class is too thin to identify. All 222 drugs yield  $\hat{\alpha}_j > 0$ , and 97% are individually significant ( $|t| > 1.96$ ); the median is 0.130 (IQR 0.087–0.226), close to the pooled  $\hat{\alpha}_0 = 0.103$ . Instrumenting each drug’s price with the rival chains’ prices cuts the wrong-sign rate from 10% under OLS to 5 drugs (2%), which are assigned the type-level mean. The pooled  $\hat{\alpha}_0 = 0.103$  and  $\hat{\sigma} = 0.393$  (headline row) enter all downstream demand-derivative calculations; the per-drug  $\hat{\alpha}_j$  give the structural model its drug-level heterogeneity.

#### 4.5 Goodness of Fit and Implied Elasticities

Three diagnostics describe how well  $\hat{\theta} = (\hat{\alpha}_0, \hat{\alpha}_1, \hat{\sigma})$  fits the data: within-nest share fit, the cross-sectional ranking of implied own-price elasticities against pharmaceutical benchmarks [Goldman et al., 2004], and a tier-2 holdout. The formulas are collected in Online Appendix A.

(A) *Within-nest share fit.* The structural within-nest share prediction (1) uses only the estimated parameters and prices—the observed share never enters the right-hand side—so it is a genuine prediction, not a mechanical contraction-mapping  $R^2$  [Conlon and Gortmaker, 2020, p. 32]. Predicted and observed within-nest shares correlate 0.73 (pre-ban), 0.86 (post-ban), and 0.78 pooled (Table 3).

(B) *Implied elasticities, benchmarked.* From the standard nested-logit formulas (2)–(3), evaluated at each drug’s pre-ban prices and shares, the median own-price elasticity is  $-1.75$  (and  $-0.6$  post-ban, at the collapsed price sensitivity and the coordinated prices, inelastic for 84% of drugs), with 99.5% of drugs negatively signed and all cross elasticities positive. The cross-class ranking agrees in sign with the U.S. Medicare estimates of Goldman et al. [2004]—chronic, essential categories price-inelastic, discretionary ones price-elastic—and the classes a chain pushes below cost are the inelastic chronic refills, not the most elastic items (Spearman  $\rho = -0.62$  across 12 ATC1 classes; Online Appendix Table 1).

(C) *Tier-2 holdout.* Refitting on tier- $\{0, 1\}$  weeks and predicting  $\hat{s}_{ijt}$  on the held-out tier-2 weeks via (1) (with  $\hat{\phi}_{ij}$  recomputed on the training subset) tests whether demand extrapolates to the fully-coordinated regime in which the welfare results of §5.4 are evaluated; the held-out correlation is 0.85.

Table 3: Demand goodness-of-fit diagnostics

|                                                               | Pre-ban | Post-ban | Pooled |
|---------------------------------------------------------------|---------|----------|--------|
| <i>(A) Within-nest share fit (1), non-circular prediction</i> |         |          |        |
| $\text{Corr}(\hat{s}_{i jt}, s_{i jt})$                       | 0.73    | 0.86     | 0.78   |
| <i>(C) Tier-2 holdout: predict on held-out tier-2 weeks</i>   |         |          |        |
| $\text{Corr}(\text{tier-2 holdout})$                          |         | 0.85     |        |

**Note:** Diagnostic (A) reports the correlation between predicted and observed within-nest shares from (1) on the pooled chain-drug-week panel; diagnostic (B) (category ranking) is in Online Appendix Table 1; diagnostic (C) reports the held-out tier-2 correlation.

Online Appendix A decomposes the pre-ban share fit further (competitive sub-regime  $\rho = 0.67$ , war episodes  $\rho = 0.88$ ): the binding constraint is the static nested-logit’s inability to capture each chain’s week-by-week selection of which drugs to push below cost, which a time-varying  $\alpha_t$  barely closes. The headline parameters suffice for the welfare calculation of §5.4, which uses tier-level prices where shares are stable.

## 5 Dynamic Structural Model

This section embeds the store-traffic value and the followers’ belief in a dynamic pricing game and asks two questions. The game is *non-stationary*: the advertising ban, the February summer holiday, and a finite detection horizon all fall inside the sample window, so it is solved by backward induction over calendar weeks rather than as a stationary recursion. First, is the estimated collapse in price sensitivity at the advertising ban (by itself, with no change in costs, patience, or any other fundamental) enough to flip the static incentive from the loss-leader war to self-enforcing coordination? Second, does the same model, rolled forward, reproduce the observed path: a price war in which coordination attempts are undercut and fail, then a post-ban wave of successful coordinations that builds to the cartel? The answer to both is yes. Beyond the estimated demand, the model adds a store-traffic value  $\mu$  and a small set of organizational, belief, and timing parameters governing how the coordination wave rolls out. Everything that does the economic work, the war/coordination flip itself, is read off the estimated demand of Section 4.

### 5.1 The dynamic game

The dynamic game has five pieces, taken in turn: the *state* (a); the per-period *payoff* and the static-Nash incentive flip that drives the war-to-cartel transition (b); the *leader* probability (c); the *follower* probability (d); and the *belief* through which the dated events enter (e).

(a) *State*. The model is built drug by drug—products are heterogeneous, not collapsed into a single representative good, and each of the  $J = 220$  drugs is sold by the three chains (each with its own demand quality  $\hat{\phi}_{ij}$ , heterogeneous across drugs), belongs to a laboratory  $L(j)$ , and carries its own war, restored, and rent prices ( $p_j^0, p_j^1, p_j^2$ ), cost  $c_j$ , and price coefficient  $\alpha_j$ , hence its own one-shot undercut gain  $G_j(\alpha_t)$ . The three tier prices and the cost are read from drug  $j$ 's own entries in the hand-coded event ledger:  $p_j^0$  is the median transacted price across its below-cost war deviations,  $p_j^1$  the median across its first ( $0 \rightarrow 1$ ) coordinations,  $p_j^2$  the median across its rent ( $1 \rightarrow 2$ ) coordinations (imputed as  $1.12 p_j^1$  for drugs that never reach the rent tier), and  $c_j$  the median implied cost (price over markup) across all of  $j$ 's events; their cross-sectional distribution, not a representative drug, enters the estimated model. Each lab carries its stake  $W_L = \sum_{j:L(j)=L} R_j$  (its drugs' total revenue). The weekly state is the tier vector  $\ell_t \in \{0, 1, 2\}^J$  (0 war, 1 restored, 2 rent), the advertising regime  $\alpha_t$ , and the calendar week.

(b) *Payoff and the static-Nash flip*. Drug  $j$ 's within-period demand is the estimated nested logit of §4.1: chain  $i$ 's share is

$$s_{ij}(\mathbf{p}_j; \alpha_j) = \frac{e_{ij}}{D_j} \frac{D_j^\omega}{1 + D_j^\omega}, \quad e_{ij} = \exp\left(\frac{\phi_{ij} - \alpha_j p_{ij}}{\omega}\right), \quad D_j = \sum_i e_{ij}, \quad \omega = 1 - \hat{\sigma}, \quad (6)$$

with  $\hat{\sigma} = 0.393$  and the three chains in one nest. Each chain carries its own estimated quality  $\hat{\phi}_{ij}$ , so the chains split the nest *asymmetrically* (Cruz Verde 0.45, FASA 0.31, Salcobrand 0.24 at the coordinated price, matching the data); the coordinated-window intercepts predict the war- and rent-period within-shares out of sample to a median 1.6 percentage points (Online Appendix A.1). The only object the ban changes is the price coefficient: each  $\alpha_j$  scales down by the common pooled-IV factor  $0.029/0.103$ . The *level* of price sensitivity is drug-specific, but the proportional collapse is taken as the economy-wide estimate because the per-drug ban interactions are individually too noisy to identify; the comparative-price ads that told shoppers who was cheapest are gone, so shares stop responding to price. Leadership, by contrast, is carried by the store-traffic asymmetry  $\mu_i$  rather than the share, so the smallest chain, Salcobrand, leads the escape (§(c), §5.3).

Beyond the drug margin, each customer is worth a store-traffic value  $\mu$ , the chain's variable profit on the rest of her basket, so the per-period payoff is

$$\Pi_{ij}(\mathbf{p}_j; \alpha_j) = M_j s_{ij}(\mathbf{p}_j; \alpha_j) (p_{ij} - c_j + \mu), \quad (7)$$

with  $M_j$  the market size and  $c_j$  the wholesale cost. Store traffic is why pricing below cost ( $p_{ij} < c_j$ ) can be rational, the lost drug margin repaid by basket profit on the extra traffic [Lal and Matutes, 1994, Thomassen et al., 2017]; held fixed across the ban,  $\mu$  lets the ban's

entire effect run through  $\alpha$ .

The transition turns on the traffic a deviation wins. A chain undercutting the coordinated price  $p^c$  to the war floor  $p^w$  while its rivals hold gains share  $\Delta s = 0.188$  with ads on ( $\alpha = 0.103$ ) but only 0.050 with ads banned ( $\alpha = 0.029$ ), a 3.8-fold collapse: with ads on a cut draws a large block from rivals, with ads banned almost none. Undercutting then pays only if the store-traffic profit on the stolen  $\Delta s$  outweighs the drug margin given up,

$$G(\alpha) = \Pi^{\text{dev}} - \Pi^{\text{coord}} = \underbrace{\mu \Delta s}_{\text{traffic profit}} + \underbrace{[s^{\text{dev}}(p^w - c) - s^{\text{coord}}(p^c - c)]}_{\text{drug-margin change} < 0}, \quad (8)$$

that is, if  $\mu$  clears a threshold  $\mu^*(\alpha) = -(\text{drug-margin change})/\Delta s$ . Because  $\Delta s$  collapses 3.8-fold while the margin term barely moves, the threshold jumps about 3.5-fold, from  $\mu_{\text{ads on}}^* = 8.6$  to  $\mu_{\text{ads banned}}^* = 30.3$ ; the estimate  $\hat{\mu} \approx 9.3$  sits just inside the lower one, so the chains fought a below-cost war with ads on, and once ads are banned the threshold rises above  $\hat{\mu}$  and the same cut turns unprofitable. The war also trapped the follower, whose alternative to matching a lift is to harvest—staying cheap while the leader is dear—drawing

$$\Delta s^{\text{harv}} = s_i(p^w, p^c, p^w; \alpha) - s_i(p^w, p^w, p^w; \alpha) = \begin{cases} 0.069 & \text{ads on} \\ 0.014 & \text{ads banned;} \end{cases} \quad (9)$$

matching beats harvesting in both regimes, but with ads on the matched price is itself undercut, so no lift bootstraps, while with ads banned undercutting is worthless and the lift holds. Operating through demand alone, the ban thus flips the deviation incentive ( $G < 0$ ) and makes coordination a one-shot best response, self-enforcing for the inelastic majority (179 of 220 drugs) without any appeal to patience or trigger strategies; the most-elastic 41 are held by the common-knowledge belief. (I test the two observed deviations, not a global static best response.)

This static flip is embedded in the full dynamic model below, carrying each drug's own gains and the two-layer organization the legal record describes.

*From daily demand to a weekly payoff.* Demand is estimated on the daily panel (§4), but the coordination game steps weekly. Chain  $i$ 's weekly payoff on drug  $j$  at tier  $\ell$  therefore scales the daily flow profit (7) up to a week,

$$\Pi_{ij}^\ell = 7 M_j^{\text{day}} s_{ij}^\ell (p_j^\ell - c_j + \mu) = N_j s_{ij}^\ell (p_j^\ell - c_j + \mu), \quad (10)$$

with  $M_j^{\text{day}}$  the daily market size and  $s_{ij}^\ell = s_{ij}(\mathbf{p}^\ell; \alpha_{j,t})$  chain  $i$ 's estimated tier- $\ell$  share under the prevailing regime, and  $N_j \equiv 7M_j^{\text{day}}$  the weekly-equivalent market.

*Value functions.* The environment is non-stationary (the ban, the February holiday,

a finite detection horizon), so each drug's continuation value is computed by backward induction over the window. For tier  $x \in \{0, 1, 2\}$ ,

$$V_j^x(t) = \Pi_j^x(\alpha_t) + \delta V_j^x(t+1), \quad V_j^x(T) = \Pi_j^x(\alpha_T), \quad (11)$$

with the weekly discount  $\delta$  fixed at 0.95 (the fit is unchanged over  $\delta \in [0.90, 0.97]$ ). Write  $\Delta V_j^1(t) = V_j^1(t) - V_j^0(t)$  for the value of holding the restored tier over the war, and  $\Delta V_j^2(t) = V_j^2(t) - V_j^1(t)$  for holding the rent over the restored tier. One object is chain-specific, and it is the structural answer to why Salcobrand is the chain least able to bear the war. The store-traffic value a loss-leader draws is itself chain-specific: each chain  $i$  has its own  $\mu_i$ , the basket profit a below-cost cut pulls into its stores, so its war value is

$$V_{ij}^0(t) = S_j^0(p_j^0 - c_j + \mu_i) + \delta V_{ij}^0(t+1), \quad (12)$$

worth  $\mu_i S_j^0$  more to chain  $i$  than the pharmacy margin alone. A higher  $\mu_i$  makes the below-cost war less painful—the basket profit offsets the drug loss—so the chain with the *lowest* store-traffic value bears the war worst, giving a chain-specific escape value  $\Delta V_{ij}^1(t) = V_j^1(t) - V_{ij}^0(t)$  that is largest for the smallest  $\mu_i$ . The estimates order the chains  $\hat{\mu}_{SB} = 8.0 < \hat{\mu}_{FA} = 9.6 < \hat{\mu}_{CV} = 10.2$ : the purely-domestic Salcobrand draws the thinnest basket and so has the biggest escape value, consistent with its cutting *least* in the war (§3.3). The in-market deviation incentives (the undercut and rent net gains) use the common  $\bar{\mu}$ ; only the leadership decision, which turns on a chain's exposure to the war, uses the chain-specific  $\Delta V_{ij}^1$ .

*Static gains, the belief-weighted future value, and Type-I EV choice.* Two one-shot gains drive each period's decision, both read off the spillover. The gain to undercutting a coordinated price,  $G_j^{01}(\alpha_t) = \mu \Delta s_j^{01} + \text{margin}_j^{01}$  of (8) (and  $G_j^{12}$  for the rent step), is the static-Nash margin: while ads are on it is positive (undercutting pays—the war) and the ban flips it negative (holding the coordinated price is the static best response—the wave); the gain to harvesting rather than following a lift,  $G_j^{\text{harv}}(\alpha_t) = \mu \Delta s_j^{\text{harv}} + \text{margin}_j^{\text{harv}}$  from the harvest spillover (9), sets the follower's static match incentive ( $G_j^{\text{harv}} < 0$  in both regimes, so a follower always prefers to match a lift). The undercut gains, weighed against the future value of coordination they forfeit—the escape value  $\delta \Delta V$ —give the net gains

$$\text{NG}_j^{01}(t) = G_j^{01}(\alpha_t) - \delta \Delta V_j^1(t+1), \quad \text{NG}_j^{12}(t) = G_j^{12}(\alpha_t) - \delta \Delta V_j^2(t+1). \quad (13)$$

(c) *Leader probability.* Leadership is an outcome of store traffic, not a free weight. The chain most willing to go first, raising price and risking a spell of being harvested, is the one with the most to gain from escaping, i.e. the chain whose below-cost war is most painful—the one with the lowest store-traffic value  $\mu_i$  (12). Under the same Type-I EV

shocks, the *first mover* of a feasible lift is chain  $i$  with the random-utility probability

$$\pi_{ij}(t) = \frac{\exp(\Delta V_{ij}^1(t)/\tau)}{\sum_k \exp(\Delta V_{kj}^1(t)/\tau)}, \quad (14)$$

the chain-specific escape value  $\Delta V_{ij}^1$  of (12) setting it (this sums to one over chains, so it fixes only who moves first). Leadership is not single-chain: each other chain  $k$  independently joins the same lift as a co-leader with probability  $\text{logit}^{-1}(\Delta V_{kj}^1/\tau + c_0)$ , rising in its own escape value, so one to three chains lead a given event and the offset  $c_0$  matches the observed co-leader rate of about 1.3 chains per event. When  $k$  chains lead, the  $3 - k$  non-leaders must each match for the lift to hold (15), so the belief enters as  $b_t^{3-k}$ , the squared  $b_t^2$  being the common single-leader case. Because  $\hat{\mu}_{SB}$  is the lowest, Salcobrand draws the least basket profit from a below-cost cut, so it bears the war worst, has the largest escape value, and leads most, while a higher-traffic rival, cushioned by the basket its cut pulls in, is a reluctant leader: the FASA record has Salcobrand “very habitually” raise first (*muuy habitualmente Salcobrand*), the others following over the next two days [Tribunal de Defensa de la Libre Competencia (TDLC), 2012]. The post-war rent reverses the order: once the war is over the *highest*-traffic chain, Cruz Verde, is the comfortable aggressor that keeps pushing prices into the rent—the  $SB \rightarrow CV$  handoff in the data. Store traffic is thus a primitive of the value function; the Salcobrand-led escape and the Cruz-Verde-led rent are both its outcome, and the within-drug symmetry means the  $\mu_i$  spread is the one channel that carries them.

The below-cost exposure corroborates the estimated  $\mu_i$  ordering out of sample. Cruz Verde bore the largest below-cost war losses (31.3% of drug-weeks below cost, the deepest median margin) and Salcobrand the smallest (25.6%, the shallowest), FASA between (§3.3)—exactly the order of the estimated store-traffic values  $\hat{\mu}_{CV} > \hat{\mu}_{FA} > \hat{\mu}_{SB}$ , because a higher store-traffic value is what makes loss-leading pay and a chain willing to go deeper below cost. That same ordering produces the leadership: the chain that loss-leads hardest, Cruz Verde, is the comfortable aggressor that leads the post-war rent, while the chain that loss-leads least, Salcobrand, is hurt most and leads the escape. The  $\mu_i$  are estimated from the leadership shares, but the exposure—an independent feature the estimation does not target—implies the same ordering, so the leadership is corroborated, not freely fit.

(d) *Follower probability.* The future value is large— $\delta \Delta V^1$  runs an order of magnitude above the one-shot gain (§5.3)—so it dominates whenever realized; but it is realized only if the coordination holds, i.e. if the followers match and no rival then undercuts. A follower’s belief  $b_t$  is exactly that probability, so it holds rather than undercuts a coordinated price iff  $b_t \delta \Delta V_j^1 > G_j^{01}$ . The continuation value is thus inside the follower’s decision, discounted



by how likely coordination is to be reached.

That belief enters twice, because coordination is a three-firm problem: the leader raises price only if it expects to be followed, and each follower matches only if it expects the other to match (a chain that lifts alone is undercut all the same). Both expectations are  $b_t$ , so a lift survives only when the leader believes, the followers believe in each other, and the hold beats the undercut. Type-I extreme-value shocks (scale  $\tau$ , fixed) smooth this into a conditional choice probability:

$$p_{ij}^\uparrow(t) = \pi_{ij}(t) \underbrace{b_t}_{\substack{\text{leader expects} \\ \text{a follow}}} \underbrace{b_t}_{\substack{\text{followers expect} \\ \text{each other}}} \underbrace{\text{logit}^{-1}\left([b_t \delta \Delta V_j^1 - G_j^{01}(\alpha_t)]/\tau\right)}_{\substack{\text{and holding beats undercutting}}}. \quad (15)$$

**The first mover** is the chain  $i$  that raises price first: which chain moves first is the random-utility weight  $\pi_{ij}(t)$ —a softmax over the chains’ chain-specific escape values  $\Delta V_{ij}^1$  (defined under “Who leads” below), so the most war-exposed chain (Salcobrand, the lowest  $\mu_i$ ) moves first most often; a first mover and any co-leaders are drawn per lab-batch when the lab organizes ((15) writes the modal one-leader case, where the two followers gate the lift at  $b_t^2$ ; with  $k$  leaders the gate is  $b_t^{3-k}$ ). **The followers**—the chains that do not lead—hold the lift iff the belief-weighted future value covers the undercut temptation ( $b_t \delta \Delta V_j^1 > G_j^{01}$ ). If it holds, a coordination. If not, a pre-ban lift is undercut ( $G_j^{01} > 0$ , undercutting still pays) and reverts, a failed attempt; in the cartel, where  $G_j^{01} < 0$ , holding strictly dominates, so the leader simply holds and retries—a non-match is not a reversion and the cartel stays clean. There is no separate down-move and no defection parameter: a coordinated drug never reverts, so the cartel holds. The only reversions in the model are the war’s failed lifts and, post-Oficio, the scrutinized rent pushes detailed below. What flips at the ban is the static-Nash margin  $G_j^{01}$ : pre-ban  $> 0$  (undercutting a coordinated price pays, so a matched lift is undercut—the war’s sporadic, mostly-failing leads), post-ban  $< 0$  (holding is the static best response—the wave), and the belief, the realization probability of the now-dominant future value, swings with it.

(e) *Belief and the dated events.* Coordination is dynamically sustainable in both regimes (§5.3); what holds the chains in the war is not that the coordinated price is unsustainable but that no one expects it to be reached, so the transition *selects* the coordinated equilibrium rather than creating it. The belief is a single Beta–Binomial posterior over a shared prior  $(a_0, b_0)$ , updated by Bayesian learning from the running tally of matched ( $S_t$ ) and reverted ( $F_t$ ) lifts; during the war it is the bare posterior,

$$b_t^{\text{war}} = \frac{a_0 + S_t}{a_0 + b_0 + S_t + F_t}, \quad (16)$$

with a low estimated prior ( $a_0/(a_0+b_0) \approx 0.02$ ): almost no one expects a lift to be matched, the few that are tried are undercut, and the pessimism is self-fulfilling. The belief enters the choice probability (15) *squared*—a lift survives only if both followers expect a follow—so for the inelastic loss-leaders ( $G_j^{01} < 0$ ) it is the only thing holding the war.

Public, dated events shift this belief; I list them once rather than re-derive each. The **binding November ban** (week 96), commonly observed, makes every firm perform the same update, adding  $\zeta \approx 283$  pseudo-successes at the onset  $t_0$ ,

$$b_t^{\text{coord}} = \frac{a_0 + S_t + \zeta}{a_0 + b_0 + S_t + F_t + \zeta}, \quad (17)$$

a shift, not a reset (the war's failures stay in  $F_t$ ), that lifts the belief in one step to about one-half—past the squared-belief ignition threshold—climbing only to about seven-tenths over the wave and never near one, so the post-ban dynamics are governed by the one-time incentive flip, not by belief continuing to rise. The **September CONAR ruling** is a non-binding precursor (no structural node); the **December onset**  $t_0 \approx 103$  is endogenous, the lag the laboratory reach-out generates; the **March FNE Oficio** raises enforcement on the rent alone (below).

The rent step reads the same belief, discounted by a factor  $e^{-\chi} \approx 0.14$  that prices the deeper push's greater fragility,

$$b_t^{\text{rent}} = e^{-\chi} b_t^{\text{coord}}, \quad (18)$$

so the rent hazard tracks the wave at a lower level and opens slowly. A rent-eligible drug—one whose own demand is inelastic enough that the higher price still pays ( $\Delta V_j^2 > 0$ )—is pushed  $1 \rightarrow 2$  in week  $t$  with probability

$$p_j^2(t) = \underbrace{b_t^{\text{rent}}}_{= e^{-\chi} b_t^{\text{coord}}} \underbrace{r_t}_{\text{scrutiny}} \logit^{-1}(-G_j^{12}/\tau) \underbrace{\mathbf{1}\{t \notin [t_0 + \delta, t_0 + \delta + \varphi)\}}_{\text{pause}}, \quad (19)$$

the discounted belief (entering once, not squared: the rent only deepens a coordination the chains already hold jointly) times a static match, scaled by the post-Oficio scrutiny factor and switched off entirely through the pause window,

$$r_t = \begin{cases} 1, & t < t_0, \\ e^{-\kappa} + (1 - e^{-\kappa}) \frac{t - t_0}{T - t_0}, & t_0 \leq t < \bar{t}, \\ 1, & t \geq \bar{t}, \end{cases} \quad \bar{t} = t_0 + \delta + \varphi. \quad (20)$$

The Oficio sets off a three-phase retreat in the rent. For the first  $\delta \approx 9$  weeks the chains keep deepening rents but under scrutiny—each match sticks only at the reduced rate  $e^{-\kappa}$ ,

lifting the rent to a plateau. They then *suspend new rent pushes* for  $\varphi \approx 9$  weeks: the rent holds, and because nothing is attempted the coordination belief is not damaged—no failed attempts enter  $F_t$ , so the cartel simply waits out the investigation. When the scrutiny lifts at  $\bar{t}$  the chains resume unobstructed ( $r_t = 1$ ) and the rent surges to its full depth.<sup>16</sup> The pause window  $(\delta, \varphi) \approx (9, 9)$  weeks is read from the institutional timeline—the Oficio (31 March) plus the chains’ observation lag—not estimated;  $\kappa$  is the one estimated scrutiny parameter ( $t_0, T$  fixed dates).

The same primitive (15) runs the pre-ban war: while ads are on  $G_j^{01} > 0$ , so any coordinated lift is undercut and reverts. Before the laboratories organize ( $t < t_0$ ) each not-yet-coordinated drug is tested directly at the follower belief itself (the leader raises only if it expects a follow),

$$\Pr\{\text{drug } j \text{ tested for a lift in week } t\} = b_t [1 + g \mathbf{1}\{t \geq \text{Aug 2007}\}], \quad t < t_0, \quad (21)$$

the most war-exposed chain (Salcobrand, lowest  $\mu_i$ ) leading and Cruz Verde’s August Desafío campaign intensifying by  $g$ ; almost every test is undercut, the reverts concentrating where  $G_j^{01}$  is largest, until the November ban flips  $G_j^{01} < 0$  and the war collapses.

*Two-layer organization.* The cartel was organized lab-by-lab through the manufacturers, the intermediaries the prosecutor identifies.<sup>17</sup> Each week a chain reaches out to a not-yet-organized lab  $L$  when the coordination it would unlock is worth more than the traffic it would forgo,

$$\Pr\{\text{chain } i \text{ organizes lab } L \text{ in week } t\} = \lambda \left[ \frac{1}{\tau_e} (b_t \delta \Delta V_{iL}^1(t) - G_L^{01}) \right]_0^1 \text{hol}(t), \quad (22)$$

summing the escape value and undercut gain over  $L$ ’s drugs. Two ordering facts the previous specification *assumed* now *emerge* from this reach-out: larger labs (a larger summed surplus) cross the threshold first, so the documented larger-labs-first sequence is an outcome; and the most war-exposed chain, Salcobrand, with the largest escape value reaches out first and leads. Engagement is one-time and bounded (active for a spell of  $w$  weeks, then disengaged); conditional on it, each drug is pushed up at its own belief-gated rate (15), drugs coordinating at staggered weeks, and a coordinated drug never reverts, so the cartel once formed holds.

<sup>16</sup>A one-time negative belief shock at the Oficio—the natural alternative—does not reproduce this plateau-then-resumption: lowering the coordination belief depresses the rent hazard *uniformly*, so the rent flattens *below* the observed plateau and never accelerates; I verified this directly. The pause holds the coordination tally intact through the watching period, so resumption produces the late climb.

<sup>17</sup>“utilizaron a los laboratorios para coordinar y monitorear el acuerdo” [Fiscalía Nacional Económica (FNE), 2008].

## 5.2 Estimation and fit

*The parameters.* The free parameters are  $\theta = (\mu_{CV}, \mu_{FA}, \mu_{SB}, \lambda, \tau_e, t_0, a_0, b_0, \zeta, \chi, \kappa, g)$ : the three chain-specific store-traffic values  $\mu_i$ , their spread carrying the leadership; the reach-out scale and temperature  $(\lambda, \tau_e)$  governing laboratory engagement (22) and its onset  $t_0$ ; the shared belief prior  $(a_0, b_0)$ ; the ad-ban pseudo-successes  $\zeta$  and the rent-tier signal penalty  $\chi$  that together micro-found the two belief shifters; the post-Oficio enforcement penalty  $\kappa$  on the rent match; and the campaign strength  $g$ —twelve in all. There is no outside option ( $\psi = 0$ ): leadership is carried entirely by the estimated  $\mu_i$  spread, whose ordering the below-cost exposure independently corroborates (above), so the split is fitted but not free. Fixed are the weekly discount  $\delta = 0.95$ , the Type-I EV choice scale  $\tau$ , the holiday factor  $hol$ , and the engagement window  $w = 26$  weeks, calibrated to the cartel’s roughly six-month documented active span rather than estimated. Relative to the earlier specification I drop the experimentation rate and the separate decline shock, let the FNE investigation raise the difficulty of the rent alone (the penalty  $\kappa$ ), and let the store-traffic value vary across chains. The per-drug objects that drive the dynamics (each drug’s undercut and rent gains  $G_j^{01}, G_j^{12}$  and shares, computed once from its own  $\alpha_j$ , prices, and cost) are inputs, not estimated here.

*Forward simulation.* For a candidate  $\theta$  I first build the per-drug value functions (11) by backward induction, then roll the weekly state  $\ell_t$  forward from January 2007 (all drugs in the war,  $\ell_0 \equiv \mathbf{0}$ ) to the end of the window. Week numbers are inherited from the full panel beginning January 2006, so the binding November ban falls at week 96 and the coordination onset  $t_0 \approx 103$  in late December 2007, about seven weeks later. Each week resolves in order: while  $t < 96$  the below-cost war cheating (21), its failed attempts the only events; the ban at week 96 both collapses price sensitivity and, as a public event, jumps the coordination belief; and from the *endogenous* onset  $t_0 \approx 103$  on—once the laboratories have organized, the lag the reach-out generates—laboratory engagement (22), the belief-gated  $0 \rightarrow 1$  coordinate step (which succeeds only if both followers match *and* every chain holds its incentive constraint, else a failed attempt that reverts), and the  $1 \rightarrow 2$  rent step for the drugs whose own demand makes it pay (post-Oficio, an unmatched rent push reverts—a coordinated drug otherwise never defects).  $B$  replications under common random numbers are averaged into the moment vector  $\hat{m}(\theta)$ , and I minimize the weighted distance to the data moments,

$$\hat{\theta} = \arg \min_{\theta} \sum_k \left( \frac{\hat{m}_k(\theta) - m_k^{\text{data}}}{w_k} \right)^2, \quad (23)$$

[McFadden, 1989, Pakes and Pollard, 1989, Bajari et al., 2007] by a multi-start derivative-free search (Powell from several random starts), chosen because the objective is a step function

of the discrete tier transitions; search and simulation use fixed seeds, so  $\hat{\theta}$  is reproducible. The hyperparameters I set rather than estimate are the replication count  $B$ , the moment weights  $w_k$ , the number of starts and the per-start iteration budget, the random seeds, and the three fixed values ( $\delta, \tau, \text{hol}$ ).

*Identification.* Each parameter maps to a distinct moment (Online Appendix Table 8). Two cross-sectional moments discipline the micro-foundations: *who cheats most*, the rank correlation between a drug’s pre-ban cheat frequency and its undercut gain  $G_j^{01}$ , pins the benefit-ordered war (21); and *who takes rent*, the  $\alpha_j$  gap between drugs that reach the rent tier and those that stop, pins the endogenous  $1 \rightarrow 2$  selectivity. The store-traffic mean  $\mu$  scales every payoff and rests on no single moment; the belief prior  $(a_0, b_0)$  and the ad-ban signal  $\zeta$  are read from the regime split of the failed attempts,  $\chi$  from the smaller and later  $1 \rightarrow 2$  count,  $\kappa$  from the post-cartel failure rebound,  $(\lambda, \tau_e, t_0)$  from the onset, peak and lab-size $\times$ timing gradient, and  $g$  from the August intensification. The  $\mu_i$  spread is identified by the leadership shares but *over-identified*: the below-cost exposure implies the same  $\mu_{CV} > \mu_{FA} > \mu_{SB}$  independently, so the split tests the mechanism.

*Estimates.* The twelve parameters are reported in Table 4 (Panel A). The mean store traffic  $\bar{\mu} = 9.3$  sits just inside the ads-on threshold above which undercutting pays, and the chains spread around it ( $\hat{\mu}_{SB} = 8.2 < \hat{\mu}_{FA} = 9.8 < \hat{\mu}_{CV} = 10.1$ ). Cruz Verde’s category financials corroborate the loss-leader economics: it sold chronic drugs below cost (chronic-drug profit  $-28$  M CLP/month) while harvesting acute ( $+2,299$ ) and non-pharma ( $+1,088$  M CLP/month) sales off the resulting traffic, a basket many times the drug loss; I do not use these aggregates to estimate  $\mu$ , since they mix drug-driven and autonomous purchases. The reach-out parameters  $(\lambda, \tau_e) = (0.61, 168)$  make engagement track the expected coordination gain, so the documented larger-labs-first sequence and Salcobrand’s lead emerge from the reach-out rather than being imposed; each organized lab is then active for the calibrated window  $w = 26$  weeks; organization begins  $t_0 \approx 103$ , about seven weeks after the November ban; the campaign strength  $g = 0.72$  reproduces the August war intensification. Firms test coordinated lifts throughout the war from January 2007 at the low follower belief, but each is undercut (the static gain  $G > 0$ ) and reverts—a failed attempt—and the belief falls further, a self-fulfilling pessimism. The shared belief starts at a low war prior  $(a_0/(a_0+b_0)) \approx 0.017$ : during the war almost no one expects a lift to be matched) and is lifted *at the November ban* by the ad-ban public signal, which adds  $\zeta = 229$  pseudo-successes—a focal common signal that, against the war’s  $\sim 129$  recorded failures, lifts the coordination belief from its low war floor to roughly three-fifths at the jump, climbing to about seven-tenths over the wave the laboratories then organize; squared across the two followers ( $b_t^2 \gtrsim 0.36$ ), and with the lab reach-out supplying many simultaneous tests, the wave fires without erasing the war history. The rent tier reads that same belief discounted by the factor  $e^{-\chi} \approx 0.14$  ( $\chi = 1.96$ ), so the supra-competitive tier stays far more selective and opens later. The model matches the record closely (Table 4, Panel B): the onset

and peak (within days), the war floor (exactly), the size-timing gradient (right sign and order,  $-0.45$  vs.  $-0.29$ ), the campaign, and the per-period failed lifts—which collapse in the cartel (near zero vs. the war’s  $\sim 129$ ) and rebound after the investigation, the signature of a cartel that holds, pauses under the investigation, then resumes deepening. It coordinates 207 of the 220 drugs. Because the coordination holds only if *every* chain prefers it—the product-firm incentive compatibility of (6), binding on whichever chain is most tempted to undercut—the model declines the score of drugs on which one chain’s undercut gain  $G_{ij}^{01}$  stays too high for the cushioned belief to hold, even as a chain’s store-traffic value cushions the still-below-cost coordinated margin enough to carry the rest. With the post-Oficio pause the rent reproduces the observed plateau-then-surge, reaching 137 of the 143 rent steps. The leadership is now an outcome of the reach-out, not a separately fitted split, and its  $\mu_i$  ordering is corroborated out of sample by the below-cost exposure: the model has Salcobrand lead 72% of the cartel wave (data 74%) and—the highest-traffic chain pushing the rent once the war is over—Cruz Verde lead 64% of the post-investigation rent (data 65%), the SB→CV handoff (Table 4). Salcobrand leads the escape and Cruz Verde the rent, the order the  $\mu_i$  spread produces through the reach-out and the exposure independently implies; it now matches both the wave share and the handoff.

Table 4: Estimated parameters and model fit

| <b>Panel A. Estimated parameters (SMM, twelve parameters)</b>      |              |              |
|--------------------------------------------------------------------|--------------|--------------|
| Parameter                                                          | Est.         | (SE)         |
| $\mu_{CV}$ store-traffic value (Cruz Verde)                        | 10.1         | (0.19)       |
| $\mu_{FA}$ store-traffic value (FASA)                              | 9.8          | (0.18)       |
| $\mu_{SB}$ store-traffic value (Salcobrand)                        | 8.2          | (0.19)       |
| $\lambda$ reach-out scale                                          | 0.61         | (0.14)       |
| $\tau_e$ reach-out temperature                                     | 168          | (41)         |
| $t_0$ organization onset (week)                                    | 103          | (0.8)        |
| $a_0, b_0$ shared belief prior                                     | 1.6, 96      | (0.25, 14)   |
| $\zeta$ ad-ban pseudo-successes                                    | 229          | (389)        |
| $\chi$ rent-tier belief penalty ( $e^{-\chi} \approx 0.14$ )       | 1.96         | (0.23)       |
| $\kappa$ FNE-scrutiny on the rent                                  | 1.97         | (0.42)       |
| $g$ Aug. campaign strength                                         | 0.72         | (0.10)       |
| <b>Panel B. Estimated model versus record</b>                      |              |              |
| Moment                                                             | Model        | Data         |
| First coordinations (Tier 0 $\rightarrow$ 1)                       | 207          | 220          |
| Rent steps (Tier 1 $\rightarrow$ 2)                                | 137          | 143          |
| Onset / peak week                                                  | 103/106      | 102/107      |
| Wave spread (SD, weeks)                                            | 4.0          | 4.6          |
| 1 $\rightarrow$ 2 lag behind 0 $\rightarrow$ 1 (weeks)             | 8.4          | 8.0          |
| <i>Coordination successes, by period</i>                           |              |              |
| war / cartel                                                       | 0/283        | 2/284        |
| post-early / post-late                                             | 30/31        | 28/49        |
| <i>Failed lifts, by period</i>                                     |              |              |
| war / cartel                                                       | 120/0        | 123/4        |
| post-early / post-late                                             | 50/13        | 46/21        |
| Below-cost war markup                                              | -0.10        | -0.10        |
| August war intensification ( $1+g$ )                               | 1.72         | 1.68         |
| <b>corr(log lab size, coord. week)</b>                             | <b>-0.45</b> | <b>-0.29</b> |
| Lab-wave SD (weeks)                                                | 5.8          | 4.9          |
| <i>Leadership (emergent from reach-out; ordering corroborated)</i> |              |              |
| SB leads the cartel wave                                           | <b>72%</b>   | <b>74%</b>   |
| CV leads the post-late rent                                        | <b>64%</b>   | <b>65%</b>   |
| SB leads the post-early failures                                   | <b>94%</b>   | <b>83%</b>   |

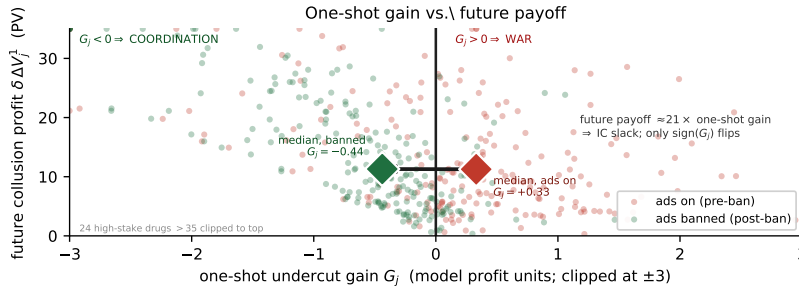
**Note:** *Panel A:* twelve parameters estimated by SMM (loss 16.9 at  $B = 240$ ; the objective adds weekly rent-path checkpoints disciplining the post-Oficio dynamics); the engagement window  $w = 26$  weeks is calibrated, fixed alongside  $\delta = 0.95$ ,  $\tau = 0.13$ , the holiday factor and  $\psi = 0$ . Standard errors: the  $\mu_i$  from a parametric bootstrap ( $B = 40$  pseudo-panels),  $\tau_e$  from the local SMM-criterion curvature, the rest carried from the one-time bootstrap. *Panel B:* each drug uses its own prices, cost,  $\alpha_j$ , lab and lab size.

*The belief transient.* Traced along the simulated path (16)–(17), the follower belief is a fast transient: low through the war, lifted by the ad-ban signal to about one-half at the jump and climbing to about seven-tenths over the wave, never near one (the war’s failures stay

in  $F_t$ ). Squared, this is already past ignition ( $b_t^2 \gtrsim 0.36$ ) once  $G_j < 0$ , so over December–March belief is *no longer the binding margin*: the wave is the mechanical consequence of the static flip, not a gradually-built belief. Belief is thus decisive for the transition—it sustains the pre-ban war and times the onset, and neither a no-belief nor a gradual alternative fits (§5.3)—but plays no role over the wave itself.

*The post-investigation rent.* The cartel does not break. The FNE’s 31 March Oficio 419 raised detection risk, yet the 220 coordinated drugs hold their price (Figure 7a); what it changes is the *deepening*, since pushing a coordinated drug to the supra-competitive rent now invites scrutiny. The chains respond by *pausing*: after a brief scrutiny-suppressed climb they suspend new rent pushes for about two months—the rent holds, and because nothing is attempted the coordination belief is undamaged—then resume once the scrutiny abates, the plateau-then-surge (Figure 7). The scrutiny enters as one penalty  $\kappa$  that, after the Oficio, lowers the probability a rent push is matched; it moves no payoff, so a reverted push is a coordination failure, not a deviation. Estimated at  $\kappa = 1.97$ , the pause reproduces the post-cartel pattern—failed lifts rebound (50 vs. near zero in the cartel) and the rent surges late to 137 of 143 (Table 4, Panel B)—and the same  $\mu_i$  spread reorders the late rent: the most-exposed chain (Salcobrand) pulls back, so Cruz Verde leads the post-investigation rent, the SB→CV handoff.

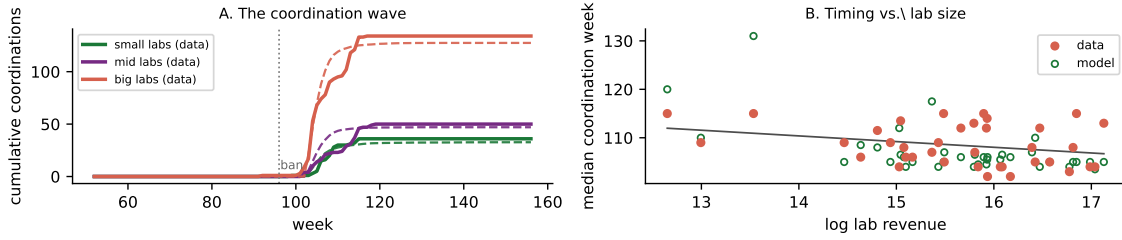
Figure 5: Coordination selected by the deviation gain



**Note:** Each drug’s one-shot gain to undercutting a coordinated price,  $G_j$  (8) evaluated on the estimated demand (horizontal), against the forgone discounted escape value  $\delta \Delta V_j^1$  (12) (vertical); red = pre-ban (ads on), green = post-ban (ads banned), diamonds = regime medians, axes clipped to  $G_j \in [-3, 3]$  and  $\delta \Delta V^1 \leq 35$  (22 high-stake drugs lie above).



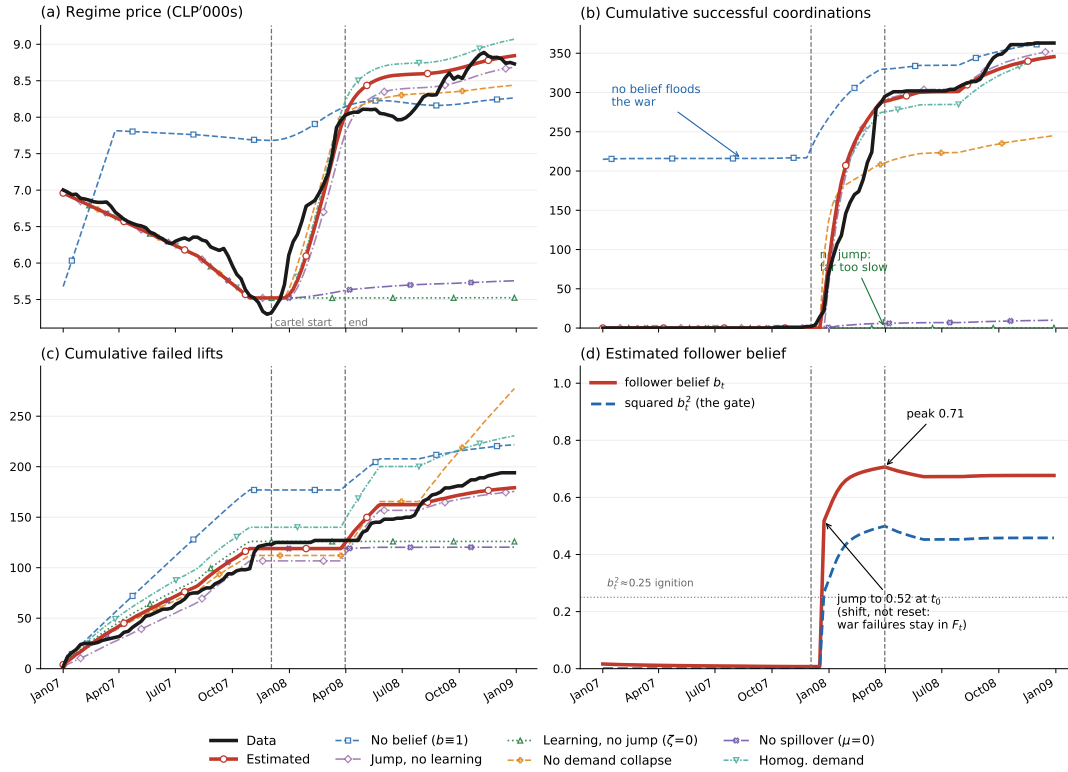
Figure 6: Model reproduces the coordination sequence



**Note:** **Panel A:** cumulative coordinations, model (dashed) vs. hand-coded data (solid), by lab-size tercile. The big- and mid-lab cohorts lead the small-lab cohort; flat through the pre-ban war, then the post-ban wave. **Panel B:** lab-level coordination week vs. log lab revenue; larger labs coordinate first (rank correlation  $-0.40$  model vs.  $-0.28$  data).

*The price path.* Figure 7a checks the model against the price series itself, with one caveat: the regime price *levels* are calibrated, not predicted—the coordinated tiers are the markup-implied prices of \$5.1 and the war floor is the pre-ban symmetric Bertrand–Nash price with store traffic, each chain maximizing  $(p - c + \mu) s(p; \alpha_{\text{pre}})$ . Since the below-cost war markup is itself a matched moment, the fitted floor reproduces the observed war price near-mechanically; the economic content is that a positive, finite  $\mu$  rationalizes a below-cost competitive equilibrium at all and then makes post-ban undercutting unprofitable. Panel (a) traces the symmetric regime path against the quantity-weighted data price—the January-2007 decline to the war floor while ads are permitted, then the recovery into the coordinated tiers once the November ban flips the incentive (the regime price holds at the floor through the war, recovers from week  $\sim 102$ , and climbs to about 9 by late 2008)—a consistency check on the whole trajectory, joined by the estimated timing, not an independent test of the level. Panels (b) and (c) make it discriminating: the cumulative wave and the failed lifts separate the estimated mechanism from the re-estimated alternatives—the belief and spillover knockouts each break a panel (the war floods, the wave is far too slow, or coordination collapses), while a misspecified homogeneous-demand spec tracks the aggregate counts but mis-shapes the wave (§5.3).

Figure 7: Structural specifications versus the data



**Note:** four panels over January 2007–December 2008. In (a)–(c) the data (solid black) are set against seven series, each re-estimated under the Pause and carrying a marker with staggered phases so overlapping curves stay distinct. The seven curves: (1) the *estimated* model (headline); and six alternatives, each switching off one ingredient—(2) *no belief* ( $b \equiv 1$ , the coordination gate always open); (3) *jump, no learning* (the ad-ban signal  $\zeta$  enters the belief but it never updates on the success/failure tally  $S_t, F_t$ ); (4) *learning, no jump* ( $\zeta = 0$ , the tally updates but there is no focal signal); (5) *no demand collapse* (the post-ban elasticity drop removed,  $\alpha \equiv \alpha_0$ , so only belief—not the static payoff—can carry coordination); (6) *no spillover* ( $\mu = 0$ , no store-traffic value); and (7) a misspecified *homogeneous* (median- $\alpha$ ) demand. The panels: (a) the regime price (quantity-weighted across drugs, model and data on the same basket; each coordinated drug’s price climbs to its tier level over about a quarter, as in the data, rather than snapping), (b) cumulative successful coordinations (the  $0 \rightarrow 1$  wave and the  $1 \rightarrow 2$  rent), and (c) cumulative failed lifts. Each belief/spillover knockout breaks one panel, while the homogeneous-demand spec tracks the aggregate counts but mis-shapes the wave; which channel each removes, and the per-moment fit, are in §5.3 and Online Appendix Table 13. Panel (c) runs above the hand-coded ledger throughout because the model counts every lift attempt while the ledger records only the salient disputes, so the discriminating feature there is the war-period steepness. (d) plots the estimated follower belief  $b_t$  and its square  $b_t^2$  (the coordination gate): the focal signal jumps it from the war floor to about one-half at  $t_0$ , and it climbs only to about seven-tenths—never near one, since the war’s failures stay in  $F_t$ —so  $b_t^2$  clears the 0.25 ignition threshold without saturating.

Because the war is now inside the model (the merged pre-ban dynamics above), the same simulation that produces the post-ban wave also produces the war's two signatures: the intensification of below-cost cheating, and the sporadic, mostly-failing coordination that preceded the cartel. While ads are permitted  $G_j^{01} > 0$ , so undercutting for store traffic pays: chains cut below cost, and a coordinated lift, even when matched, is undercut, so it reverts. The below-cost war is broad from the January-2007 decline onset and intensifies as Cruz Verde's August-2007 "Desafío" campaign (21) escalates the advertising arms race, peaking in the weeks before the ban; once the ban flips  $G_j^{01} < 0$  the cheating unwinds and coordination becomes self-enforcing. The weekly below-cost deviation series (its pre-ban level, its August intensification, and its post-ban collapse) is matched through the war-series and war-floor moments of Online Appendix Table 8. Coordination meanwhile is sporadic and reverts throughout the war (the cumulative coordinated stock stays near zero until the ban, then rises along the wave) which is just the high war-window failure rate of Online Appendix Table 7. The mechanism thus reproduces the whole episode (escalating loss-leader war, failed coordination, ban, self-enforcing cartel) from one store-traffic primitive and the regime flip in  $G_j^{01}$ . The estimated campaign strength  $g = 0.72$  reproduces the August intensification ( $1.58\times$  vs. the data's  $1.68\times$ ), and the cheating collapses once the ban flips  $G_j^{01} < 0$ .

The game is non-stationary (the ban, the holiday, and a finite horizon set by detection), so a forward-looking firm's problem is the finite-horizon dynamic program already solved by the value functions (11). Their structure sharpens the interpretation. With patient firms and grim-trigger punishment the sustainability constraint is slack in both regimes—the discounted cooperation premium exceeds the one-shot undercut gain by more than an order of magnitude at the median drug (Figure 5), and the constraint binds for none of the 220—so coordination would be dynamically sustainable even before the ban. The war→coordination transition is therefore not a change in what is sustainable but a change in what is a static best response: pre-ban the war is the stage-game Nash ( $G > 0$ , undercutting pays), post-ban coordination is ( $G < 0$ ). Coordination is dynamically sustainable in both regimes, so the ban *selects* the coordinated equilibrium rather than making one newly possible; what it changes is the *static* incentive. Pre-ban the coordinated price is not a static Nash at all (every chain gains by undercutting it,  $G > 0$ ), so it could be sustained only as punishment-supported collusion—which needs the trigger strategies the chains, as the failed war-period attempts show, did not in fact coordinate on. The ban flips the undercut incentive to  $G < 0$ , making the coordinated price a static best response—self-enforcing, needing no trigger strategies or trust—and the common-knowledge belief jump of the previous paragraph shifts expectations onto it; the ban supplies both, and so selects the cooperative equilibrium. The static flip holds for 179 of the 220 drugs (81%); the other 41—the most elastic, whose undercut gain  $G_{ij}$  stays positive for at least one

chain even post-ban—are not a static Nash, and the ban selects them through the belief shift alone, as dynamically sustainable coordination. Two consequences follow. First, for the 179 static-Nash drugs, because post-ban coordination is a one-shot best response *for every chain* it holds at every date including the terminal one; backward induction yields no finite-horizon unraveling, unlike punishment-sustained collusion. Second, for the 179 static-Nash drugs the observed failures to match are not incentive-constraint violations (undercutting is myopically unprofitable post-ban) but the idiosyncratic Type-I EV shocks of (13), infrequent and dispersed across all three chains; the 52 non-static-Nash drugs, where undercutting still pays for some chain, supply the rest. That the model’s static-Nash selection flags 52 non-static-Nash drugs while only a score fail to coordinate, and pure dynamic enforcement would flag none, places the truth between the two—partial enforcement—and is why I read that count as corroborating, not identifying.

### 5.3 Identification and robustness

The structural exercise makes one sharp point—the advertising ban, acting only through the estimated collapse in price sensitivity, is sufficient to turn the loss-leader war into self-enforcing coordination—and a set of re-estimations shows it does not rest on a tuned parameter. Each check re-estimates or re-simulates the model with one ingredient switched off and is reported against the data in Figure 7 and Online Appendix Tables 13–11.

*Robustness checks.* Belief is essential and dated to the binding November ban: a no-belief model floods the war (217 pre-ban coordinations against the data’s 2), and without the focal jump the wave *never ignites* inside the sample—a learning-only belief ( $\zeta = 0$ ) coordinates essentially no drugs and fits an order of magnitude worse ( $\sim 2300$  against the headline 16.9). Forcing the jump to the September CONAR ruling instead fits about 44× worse (loss 751, the onset pulled to week 88 and the war flooded), so the data identify a single shift at the binding November event. The focal jump is the *dominant* channel: a jump-only belief that drops the Bayesian tally fits nearly as well as the headline (19.3 against 16.9), so the gradual learning adds little once the jump is in—the earlier impression that learning contributed materially was an artifact of the pre-pause model, which leaned on a rising belief to fit the post-investigation rent that the pause now handles directly. The static flip needs the spillover: with  $\mu = 0$  the below-cost war is no longer an equilibrium (the war floor turns positive), and with a uniform store-traffic value the  $\mu_i$  spread is gone and Salcobrand’s leadership collapses to chance. The flip is not a knife edge: the post-ban threshold  $\mu^*$  jumps about 3.5× at the estimated  $\alpha$ -drop and the ratio  $\mu_{\text{post}}^*/\mu_{\text{pre}}^*$  stays in  $[3.3, 3.8]$  under  $\pm 10$ –30% perturbations of wholesale cost and  $\sigma$ . The model is not a fully solved Markov-perfect equilibrium with endogenous timing; it is a transparent mapping from the estimated demand collapse to the static flip, and with the post-Oficio pause it now reproduces the late rent (137 of 143 rent steps).

*An outside option instead of chain-specific store traffic.* The leadership does not hinge on the store-traffic parameterisation. Re-estimating with a common  $\mu$  and three firm-side war-loss cushions  $\psi_i$  (the Editor’s earlier reading) reproduces the Salcobrand-first split but fits less well (loss 31.8; Online Appendix H, RC4): any chain asymmetry that makes the below-cost war differentially painful orders the chains the same way, but the store-traffic spread fits better, is the *same* primitive that already rationalises the war, and has its ordering corroborated out of sample by the below-cost exposure ( $\mu_{CV} > \mu_{FA} > \mu_{SB}$ ), whereas the  $\psi_i$  magnitudes are pinned by no chain financial.

*If the demand estimate is wrong.* Because the whole flip is read off the estimated demand, I re-estimate the structural model under three demand specifications (Online Appendix Table 10): the headline per-drug rival-IV  $\hat{\alpha}_i$  (loss 16.9), the per-drug OLS  $\hat{\alpha}_i$  (15.9), and a homogeneous demand at the median  $\hat{\alpha}$  (30.9). The two heterogeneous specifications fit within about a point of each other and reproduce the leadership; homogeneous demand fits far worse because, without the per-drug heterogeneity, the wave is too concentrated and the rent too slow—so the per-drug elasticity dispersion, not a knife-edge level, is what spreads the wave and times the rent.

## 5.4 Welfare of the equilibrium transition

What did the transition cost consumers? Because the war was below-cost and the cartel a coordinated increase, the answer depends on the counterfactual. I evaluate the settled coordinated prices of January–April 2008 (fourteen weeks, the back of the nineteen-week wave) against two benchmarks: the below-cost war prices of mid-2007 (what consumers faced when the cartel formed) and the pre-war 2006 baseline, computing for each drug the consumer transfer and, from the nested logit, the deadweight loss from the induced quantity distortion.<sup>18</sup>

The measure is *within the pharmaceutical market*—the drug overcharge and the quantity-distortion deadweight loss, both identified by the estimated drug demand. I do not extend it to the non-pharmaceutical basket, which the case record cannot price; the coordination raised *drug* prices, so this counts the pharmaceutical overcharge alone, a clean and conservative measure of what the transition cost drug buyers.

The coordinated path extracted supra-competitive rents on many products (Table 5): within the cartel window 37% of the coordinating drugs reached Tier 2 rent extraction—the rent deepened to 143 drugs (65%) by year-end as it ratcheted post-Oficio—priced at a

<sup>18</sup>Per drug, with the estimated nested logit ( $\sigma = 0.393$ ; post-ban  $\alpha = 0.029$ , the same demand the structural model uses). The consumer-surplus change between the coordinated price  $p^c$  and benchmark  $p^b$  is  $M [\text{CS}(p^b) - \text{CS}(p^c)]$ , with  $\text{CS}(p) = \alpha^{-1} \log(1 + [N e^{(\phi - \alpha p)/(1 - \sigma)}]^{1 - \sigma})$ , decomposed into the transfer  $(p^c - p^b) \times$  quantity and the residual deadweight loss;  $M$  is backed out from the cartel-window quantity and the observed 0.92 nest share; here  $N = 3$  counts the inside chains, not the market size  $N_j$  of (4), which enters only through  $M$ .

median 1.48× wholesale cost and 1.11× the 2006 baseline (11% above it)—the margins the conduct was fined for [Fiscalía Nacional Económica (FNE), 2008, Tribunal de Defensa de la Libre Competencia (TDLC), 2012].

Table 5: Welfare of the coordinated price path

|                                                       |                      |
|-------------------------------------------------------|----------------------|
| <b>Panel A. Coordinated prices above margin</b>       |                      |
| Drugs reaching Tier 2 within the window               | 37%                  |
| Median Tier-2 price / wholesale cost                  | 1.48×                |
| Coordination events priced > 1.4× cost                | 43%                  |
| Median Tier-2 price / 2006 baseline (% above)         | 1.11× (11%)          |
| <b>Panel B. Consumer harm over the window</b>         |                      |
| Transfer vs. below-cost war (% of sales)              | +CLP 3.4bn (+11%)    |
| of which from the Tier-2 rent products                | 80%                  |
| Gross rent vs. 2006 baseline (above-historical drugs) | +CLP 1.1bn           |
| Deadweight loss (quantity distortion)                 | <CLP 0.03bn (< 0.1%) |

**Note:** Over the settled fourteen-week window (Jan–Apr 2008, three chains; window sales ≈CLP 30.7bn). Panel A: how far coordinated prices stood above margin. Panel B: consumer harm, the overcharge on the transacted quantity (transfer) plus the deadweight loss from the quantity response in the estimated nested logit.

Relative to the below-cost war prices consumers had been paying, the coordinated path cost them about 11% (CLP 3.4bn), four-fifths of it from the Tier-2 rents—the order of magnitude in the enforcement record (the FNE’s gross-revenue estimate of ~CLP 27bn, the ~US\$38M fines upheld by the Supreme Court, and the ~CLP 2.1bn later paid in consumer compensation; Fiscalía Nacional Económica (FNE), 2008, Tribunal de Defensa de la Libre Competencia (TDLC), 2012). The deadweight loss is small (<CLP 0.03bn, under 0.1% of sales): post-ban demand is inelastic, so the increase moved little quantity and the harm is overwhelmingly a transfer. The extraction was selective—the cartel held the high-volume, salient staples near competitive levels (volume–overcharge correlation  $-0.14$ ) while taking rents on the less-salient products—so its bite is distributional, concentrated in the rent products and layered on the elimination of the war’s below-cost subsidy.

## 6 Conclusion

I have analysed the equilibrium transition from loss-leader competition to coordinated pricing among the three large Chilean retail pharmacy chains in 2007–2008. Three findings support the analysis.

First, the transition is grounded in demand. Comparative-price advertising made

the cheapest chain's below-cost loss-leader pricing rational by drawing store traffic; the estimated price coefficient collapses from 0.103 to 0.029 when that advertising is banned, so undercutting no longer wins the traffic. This single estimated change flips the one-shot incentive to undercut from positive (war) to negative (coordination). Backward induction shows the flip does not change what is sustainable but what is self-enforcing: the coordinated price is not a static equilibrium with ads on (undercutting pays) and becomes one with ads banned, so it needs no patience or trigger strategy and the finite horizon does not unravel it.

Second, a full structural model with the heterogeneity of all 220 coordinating drugs and a two-layer (laboratory, then drug) organization reproduces the observed path: the failed-attempt war, the post-ban wave with its February-holiday pause, the division between margin restoration and rent extraction, the Salcobrand leadership documented in the case file, and, as a test of the organization the legal record describes, the sequence in which the larger laboratories coordinate first. Post-ban failures, which are unmatched lifts and reverted rent pushes rather than defections from already-coordinated prices, arise at a positive frequency across all three chains as idiosyncratic shocks, not incentive-constraint violations.

Third, the welfare cost is a transfer, not an efficiency loss. The coordinated path extracted supra-competitive rents on a large share of products, 37% reached rent-extraction prices near 1.5× wholesale cost, costing consumers about CLP 3.4bn over the cartel window relative to the prices they had been paying, four-fifths of it on the rent products. Because post-ban demand is inelastic the deadweight loss is negligible; the cartel's bite is distributional and concentrated where consumers were least attentive.

The model's economic content rests on objects estimated or documented outside it (the demand system, the wholesale costs and tier prices, and the laboratory structure and leadership from the legal record) with the store-traffic value bounded by the chains' own below-cost behaviour and a handful of organizational-timing parameters calibrated by simulated method of moments. The gradual, four-month rollout, which the prosecutor's complaint describes as expanding as the agreement's success was verified,<sup>19</sup> is reproduced by the endogenous laboratory reach-out—more chains find it worth organizing a laboratory as the coordinated price becomes a static best response and beliefs shift, drug by drug—rather than by a separate contagion term; what gradual belief-learning adds beyond this is a secondary refinement (§5.3), and disentangling it from antitrust-detection risk and operational complexity is a useful direction for future work.

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<sup>19</sup>Original: "a medida que se verificaba el éxito del acuerdo" [Fiscalía Nacional Económica (FNE), 2008].

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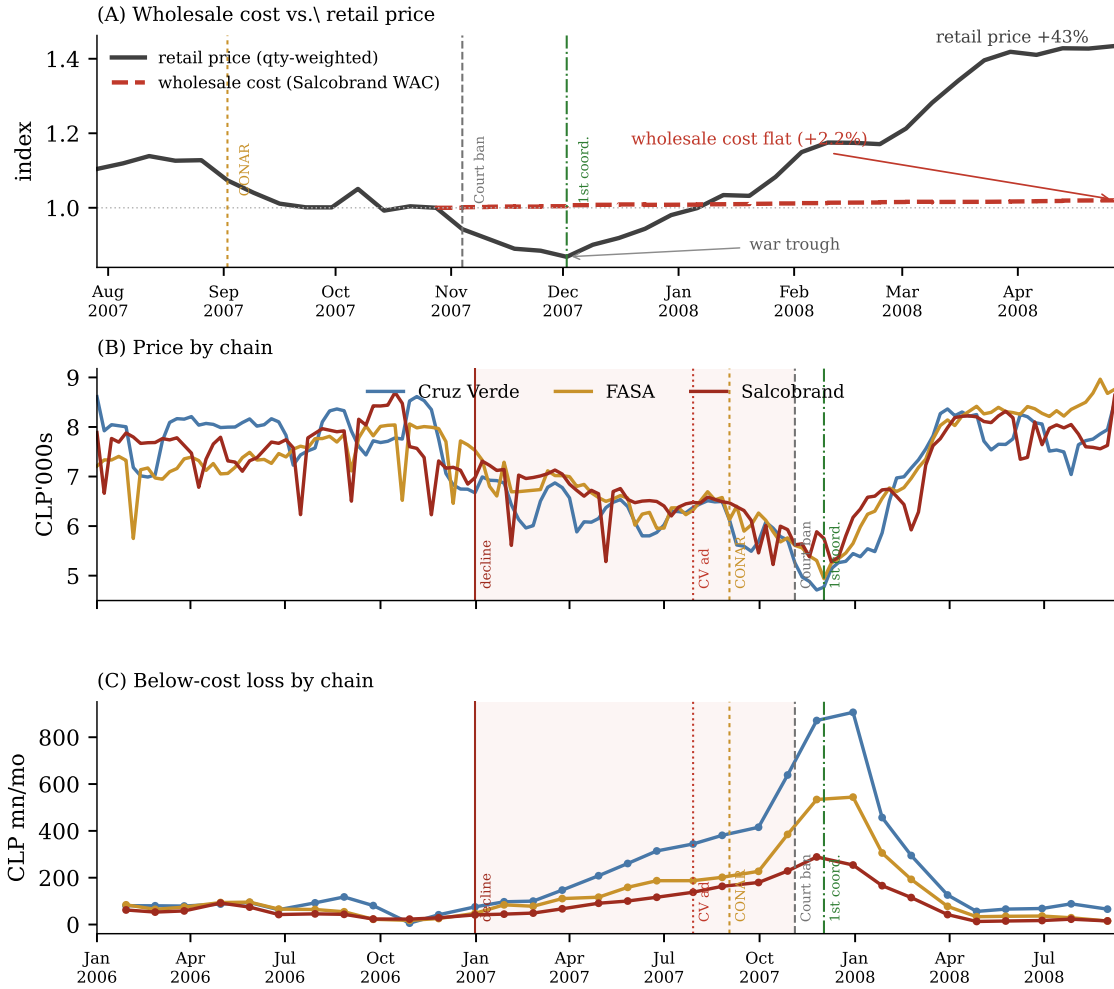
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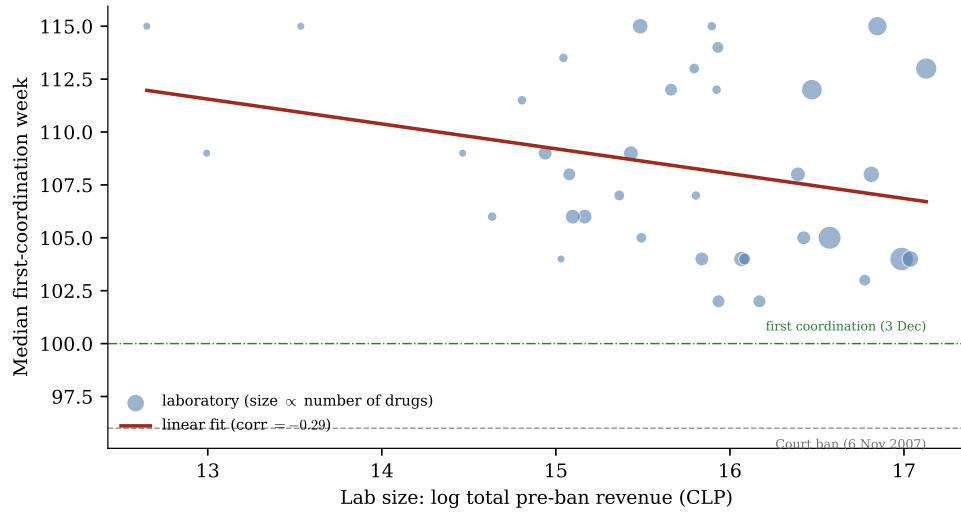
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Figure 1: Wholesale cost, prices, and below-cost loss



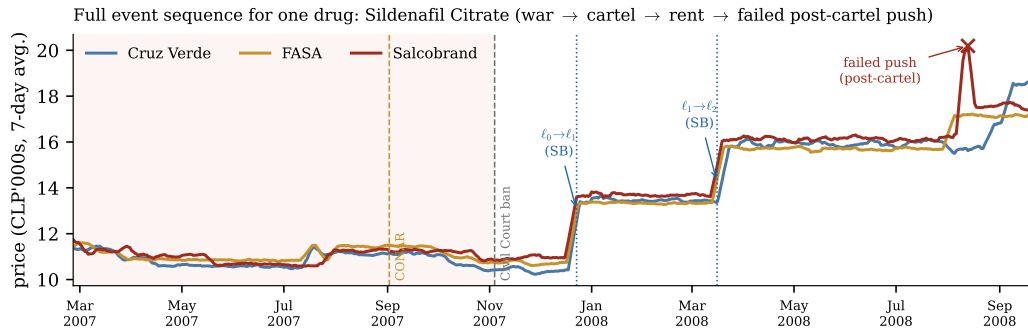
**Note:** (A) Salcobrand wholesale-cost index (dashed) and quantity-weighted retail-price index (solid), both indexed to the cost-series start, November 2007–May 2008. (B) Weekly quantity-weighted average price by chain across the 222 hand-coded drugs, 2006–2008. (C) Monthly below-cost revenue shortfall by chain,  $\sum_j \max(c_j - p_{ijt}, 0) q_{ijt}$ . The CONAR self-regulatory ruling (7 September 2007) and the binding Civil Court order (6 November) are marked in Panels B–C.

Figure 2: Larger laboratories coordinate first



**Note:** Each point is one laboratory (37 labs, 221 coordinated drugs), at its size and the median week of its drugs' first three-chain coordination. *Lab size* is the laboratory's total pre-ban revenue (log CLP), summed across its drugs—the stake  $W_L$  the structural model assigns it. Larger labs coordinate earlier (correlation  $-0.29$ ; the structural model reproduces this at  $-0.38$ , Table 4). Point area is proportional to the laboratory's number of drugs. Dashed and dash-dotted lines mark the binding 6 November court ban and the first sustained coordination, 3 December 2007.

Figure 3: Example coordination: Sildenafil price path



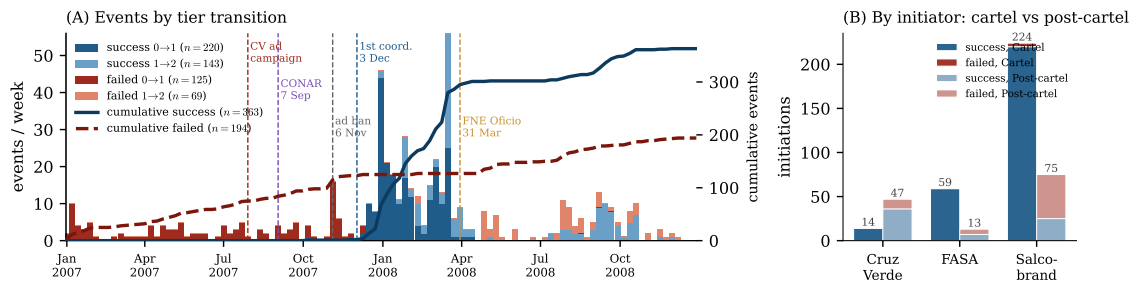
**Note:** Weekly chain price for Sildenafil Citrate (Drug #80) with the hand-coded tier transitions marked. The three chains move together within the same week at each tier step; Salcobrand initiates both.

Table 1: Coordination attempts outside the cartel window

| <b>Panel A. Principal lab-batch calls, Oct–Dec 2007 (event-level)</b>                   |                   |       |            |                            |
|-----------------------------------------------------------------------------------------|-------------------|-------|------------|----------------------------|
| Dates                                                                                   | Manufacturer      | Drugs | Initiator  | Outcome                    |
| Oct 29–Nov 4                                                                            | Pfizer            | 6     | Salcobrand | Failed                     |
| Nov 5–13                                                                                | Laboratorio Chile | 17    | Salcobrand | Failed                     |
| Nov 12–18                                                                               | Roche             | 6     | Cruz Verde | Failed                     |
| Nov 12–18                                                                               | Merck             | 4     | FASA       | Failed                     |
| Dec 3–9                                                                                 | Andromaco         | 5     | Salcobrand | Partial (2/5)              |
| Dec 10–16                                                                               | Tecnofarma        | 4     | FASA       | Success                    |
| Dec 10–16                                                                               | Bago              | 3     | Salcobrand | Success                    |
| Dec 17–23                                                                               | Saval             | 5     | Salcobrand | Success                    |
| <b>Panel B. Post-investigation rent batches, Apr–Dec 2008 (event-level, Tier 1 → 2)</b> |                   |       |            |                            |
| Dates                                                                                   | Manufacturer      | Drugs | Lead chain | Outcome                    |
| Jul 30                                                                                  | Pfizer            | 2     | Salcobrand | Rent (single-chain)        |
| Sep 1                                                                                   | Pfizer            | 3     | FASA       | Rent (Cruz Verde followed) |
| Sep 7                                                                                   | Boehringer Ing.   | 2     | Salcobrand | Rent (single-chain)        |
| Sep 11                                                                                  | Novartis          | 2     | Cruz Verde | Rent (single-chain)        |
| Sep 14                                                                                  | Boehringer Ing.   | 2     | Cruz Verde | Rent (single-chain)        |
| Sep 14                                                                                  | Pfizer            | 4     | Cruz Verde | Rent (single-chain)        |
| Sep 21                                                                                  | Sanofi-Aventis    | 2     | FASA       | Rent (Cruz Verde followed) |
| Oct 15                                                                                  | GlaxoSmithKline   | 2     | Cruz Verde | Rent (single-chain)        |

**Note:** Coordination attempts outside the cartel window. *Panel A:* Principal lab-batch calls, October–December 2007. A “batch call” is a week in which a chain raised prices on  $\geq 3$  drugs from the same manufacturer simultaneously; columns show the initiating chain and three-chain outcome (“Failed” = initiating chain reverted with no rival follow; “Partial” / “Success” = at least one / all rivals followed and held). *Panel B:* Post-investigation rent batches, April–December 2008, after the FNE’s official notice No. 419 on 31 March 2008: multi-drug Tier-1 → 2 increases on drugs the cartel had already coordinated. New (Tier-0 → 1) coordination had stopped, but rent extraction continued: eight such batches held (two multi-chain, the rest a lead chain ratcheting its own coordinated drugs), while 127 broad single-chain re-coordination lifts failed with no rival follow (SB 82, CV 29, FASA 16). The *Drugs* column is each batch’s size; the lab-level size gradient—larger laboratories coordinate first—is shown in Figure 2.

Figure 4: Coordination events over the episode



**Note:** Coordination events over January 2007–December 2008, decomposed two ways. Over the episode the ledger records 363 successful coordination steps (220 first-tier  $0 \rightarrow 1$ , 143 rent  $1 \rightarrow 2$ ) against 194 failed or reverted attempts (125 at  $0 \rightarrow 1$ , 69 at  $1 \rightarrow 2$ ). **Panel A:** weekly events split four ways—successful  $0 \rightarrow 1$  (dark blue) and  $1 \rightarrow 2$  (light blue) versus failed  $0 \rightarrow 1$  (dark red) and  $1 \rightarrow 2$  (light red); the right axis plots cumulative successes (solid) and failures (dashed). (Failed attempts carry no recorded tier; I assign each to  $0 \rightarrow 1$  or  $1 \rightarrow 2$  by whether the drug had already coordinated by the attempt date.) **Panel B:** initiations by chain, each split successful versus failed and cartel-window versus post-cartel. Vertical dashed lines (chronological): Cruz Verde’s August comparative-price campaign, the 7 September CONAR ruling, the 6 November ban, the 3 December first coordination, and the 31 March FNE official notice No. 419.



**Online Appendix** (for online publication; available at the journal website). The supplement collects: Appendix A, demand-fit decomposition; Appendix B, hand-coded events versus the official cartel indicator; Appendix C, market-size robustness; Appendix D, data construction and the batch-call ledger; Appendix E, the structural estimation sample and targeted moments; Appendix F, robustness to the demand specification; Appendix G, structural robustness checks; Appendix H, example war attempts; Appendix I, the regulatory timeline and case outcome.